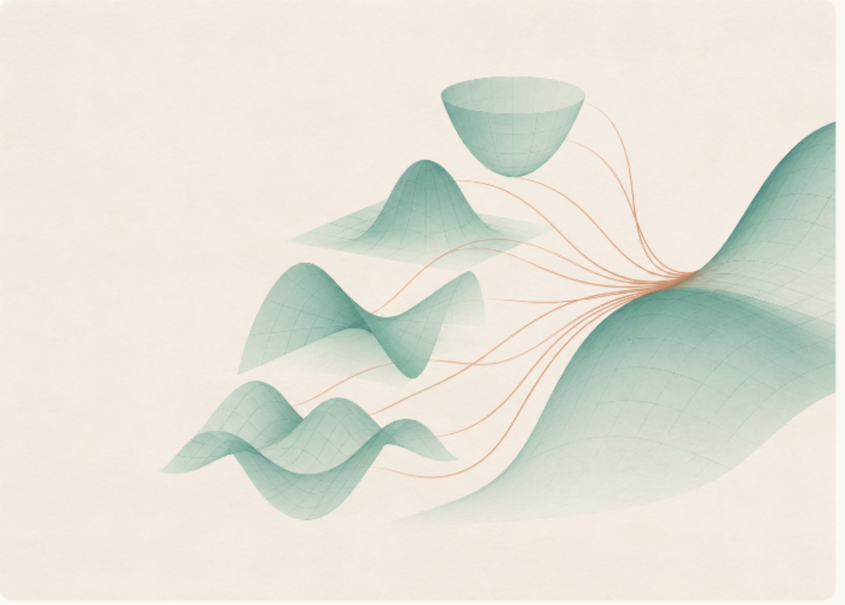
How machines learn

A delivery-time predictor

Saad Anjum, CTO

11 November 2026



An order arrives

The customer is 2 km from the restaurant.

How long will delivery take?

Distance is an input supplied by the order.

Preparation takes time

Our first preparation-time guess

3 minutes

An adjustable estimate: a parameter

Travel takes time

Our first travel-time guess

2 min / km

Travel takes time

Our first travel-time guess

2 min / km

For this 2 km order: **4 minutes**

Our first prediction

3 minutes preparation + 4 minutes travel

Estimated delivery time

7 minutes

The delivery finishes

Observed delivery time

11 minutes

The same 2 km order / synthetic teaching example

Feedback from several deliveries

Synthetic teaching records

Distance	Observed delivery time
1 km	8 minutes

Feedback from several deliveries

Synthetic teaching records

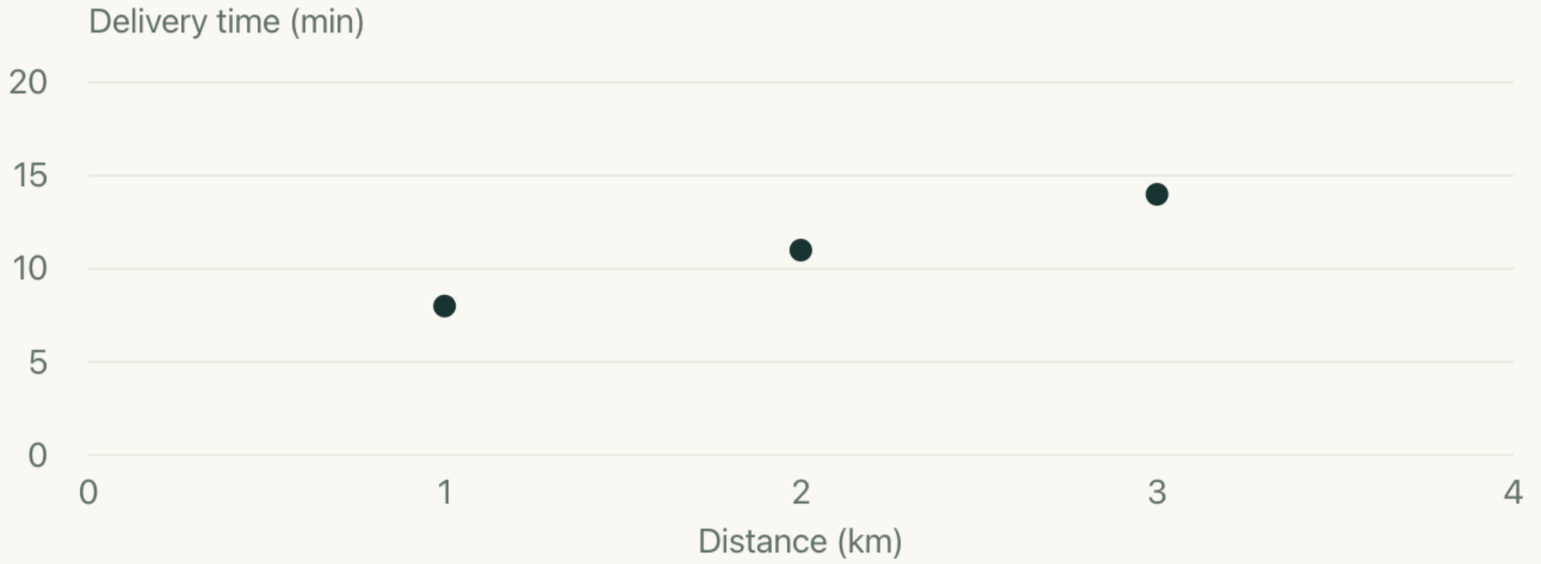
Distance	Observed delivery time
1 km	8 minutes
2 km	11 minutes

Feedback from several deliveries

Synthetic teaching records

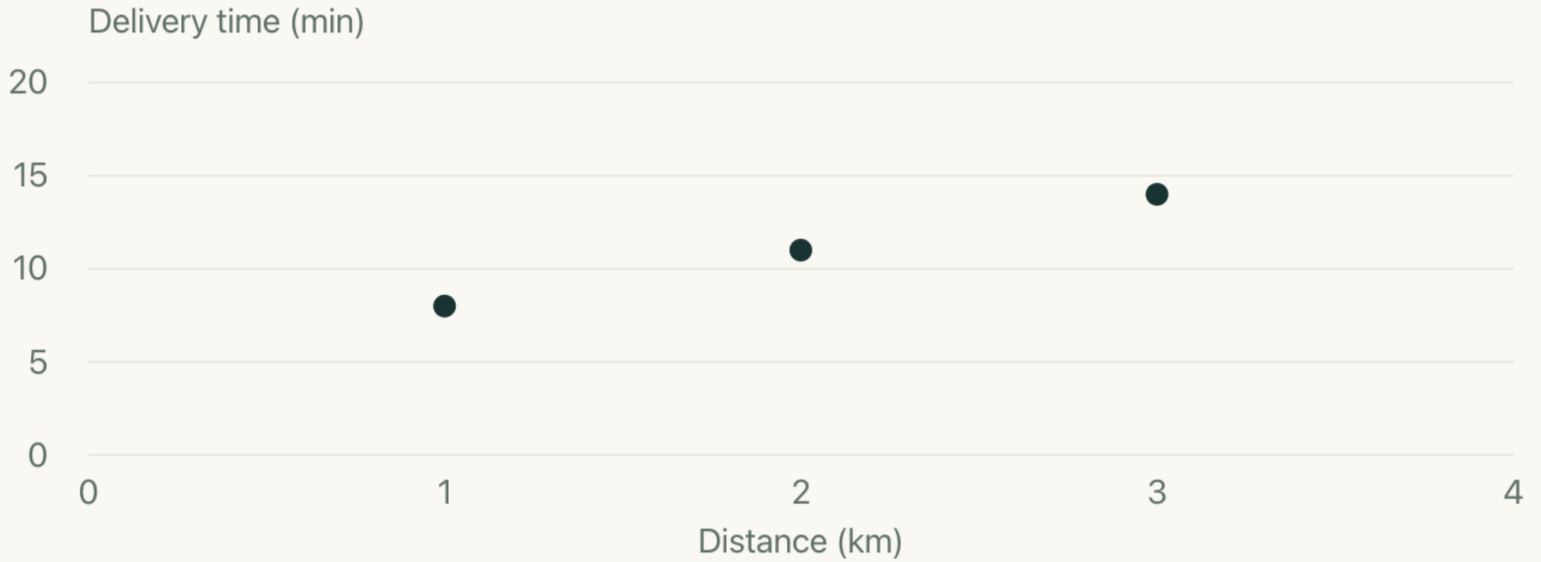
Distance	Observed delivery time
1 km	8 minutes
2 km	11 minutes
3 km	14 minutes

Our observations on a graph



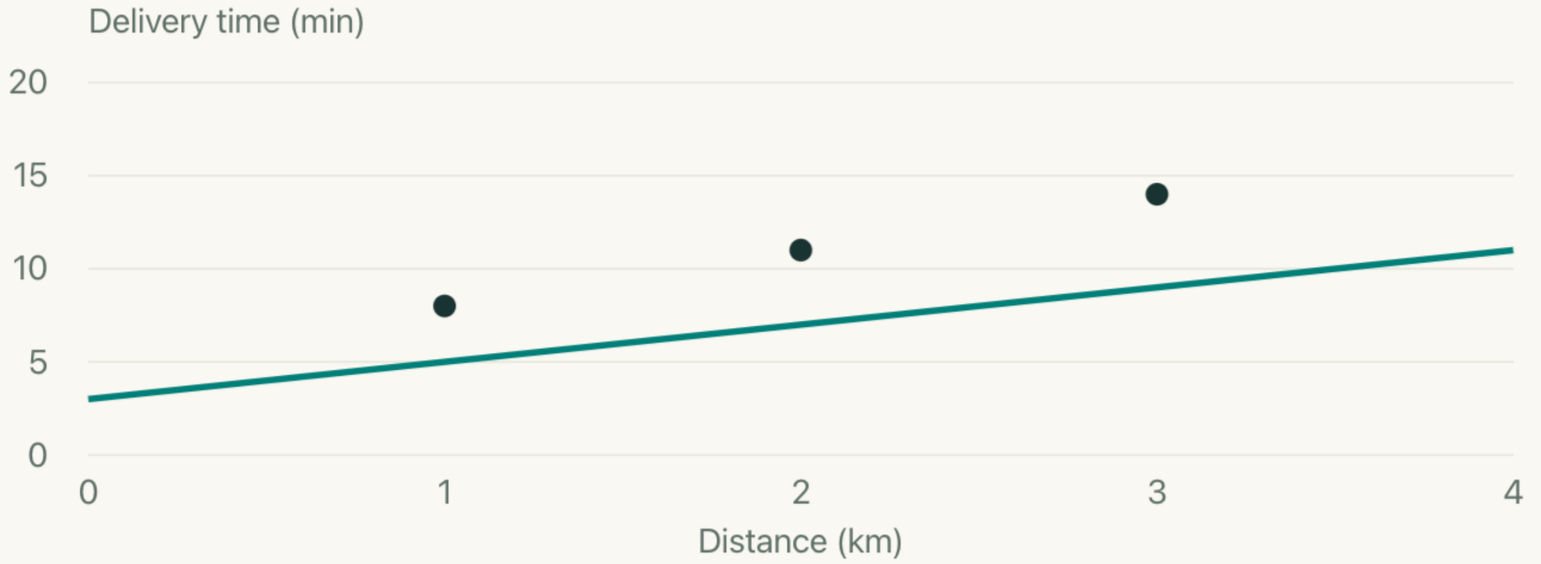
Synthetic delivery observations

Our model makes a line



The same observations, with our first prediction rule

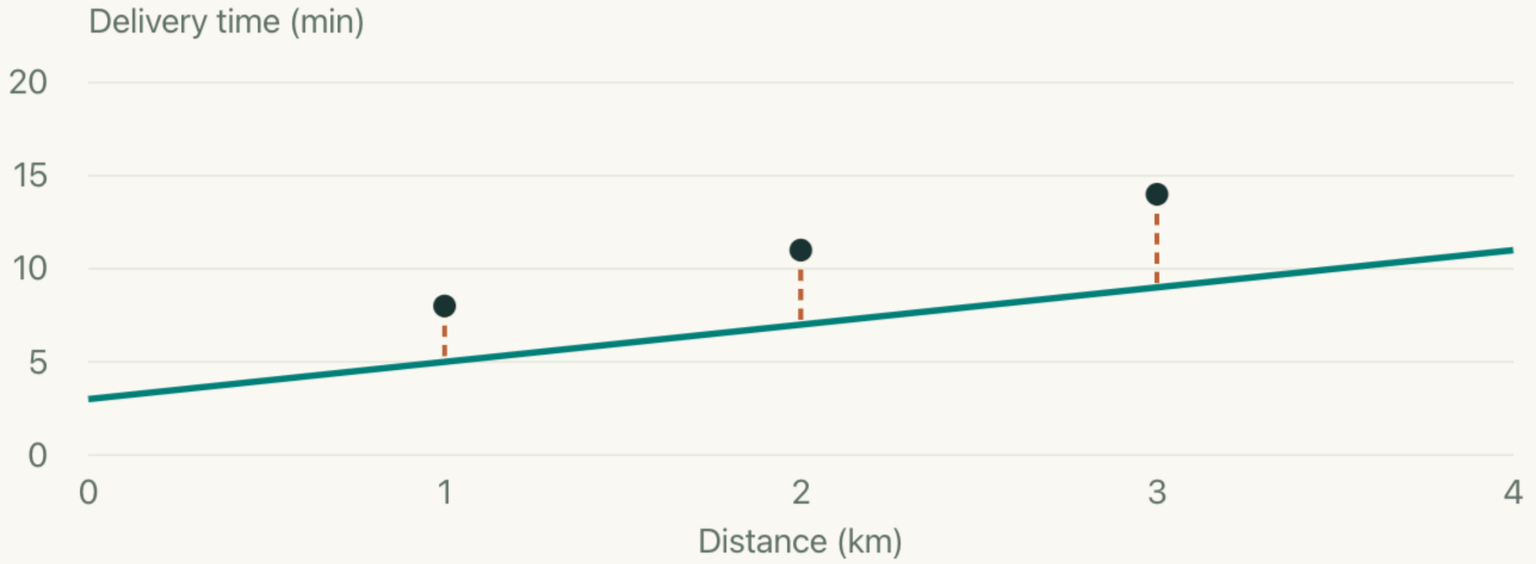
Our model makes a line



The same observations, with our first prediction rule

```
Estimated minutes = 3 + distance × 2
```

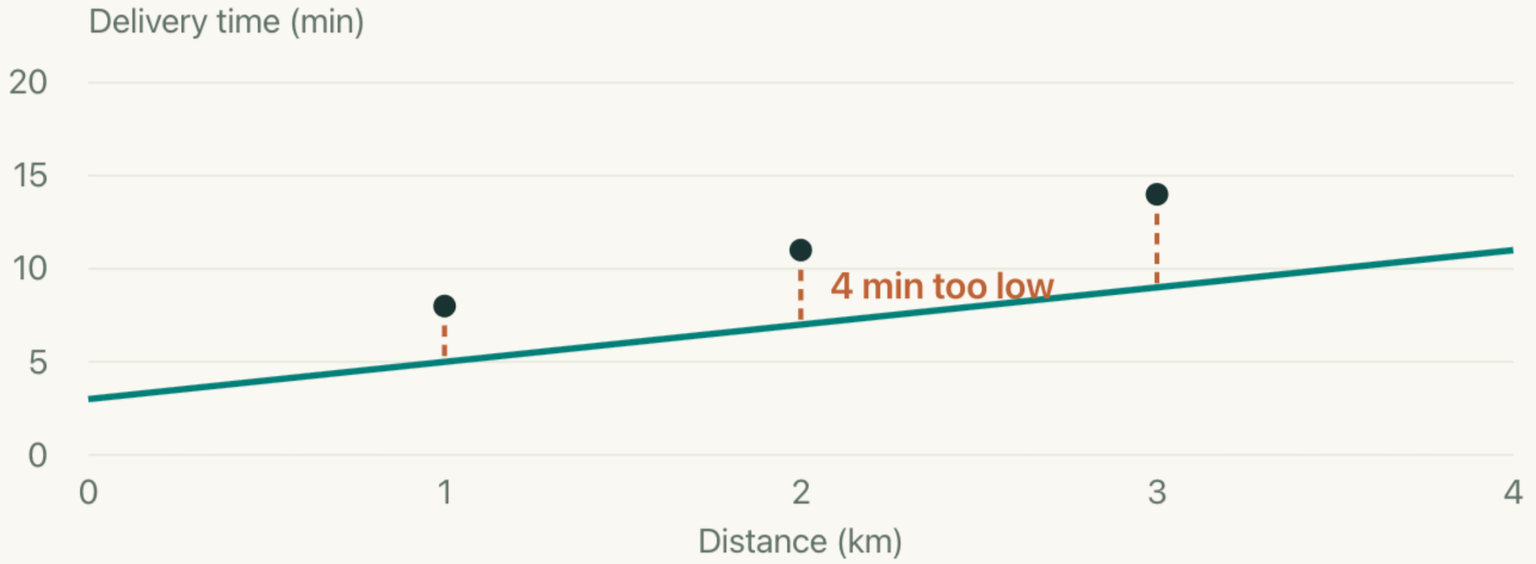
Our model makes a line



The same observations, with our first prediction rule

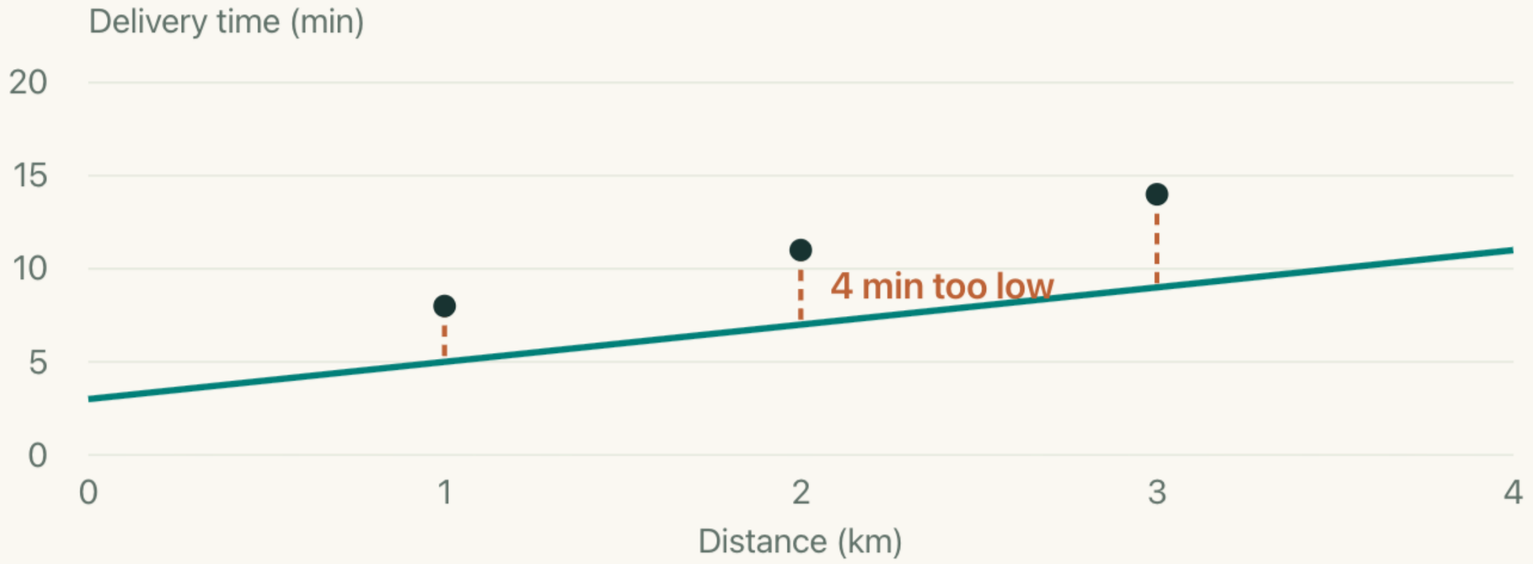
```
Estimated minutes = 3 + distance × 2
```

Measuring the mismatch



Each vertical gap is a **prediction error**.

Measuring the mismatch



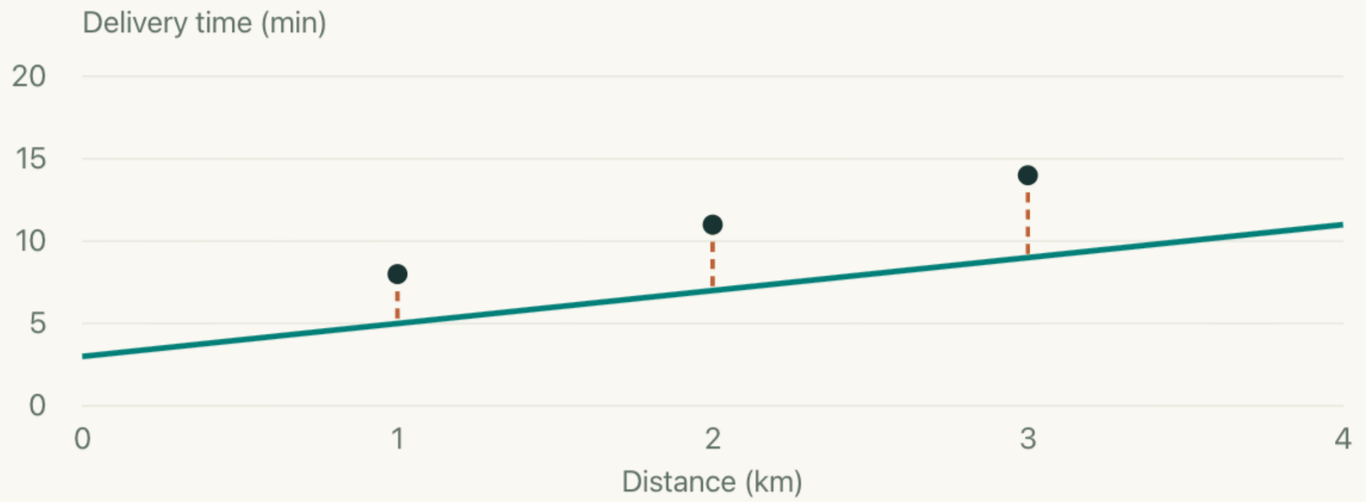
Each vertical gap is a **prediction error**.

Loss is a score calculated from those errors.
Smaller is better.

Tuning our two parameters

`Estimated min = 3.0 + distance × 2.0`

Loss (MSE, min²) **16.6667**



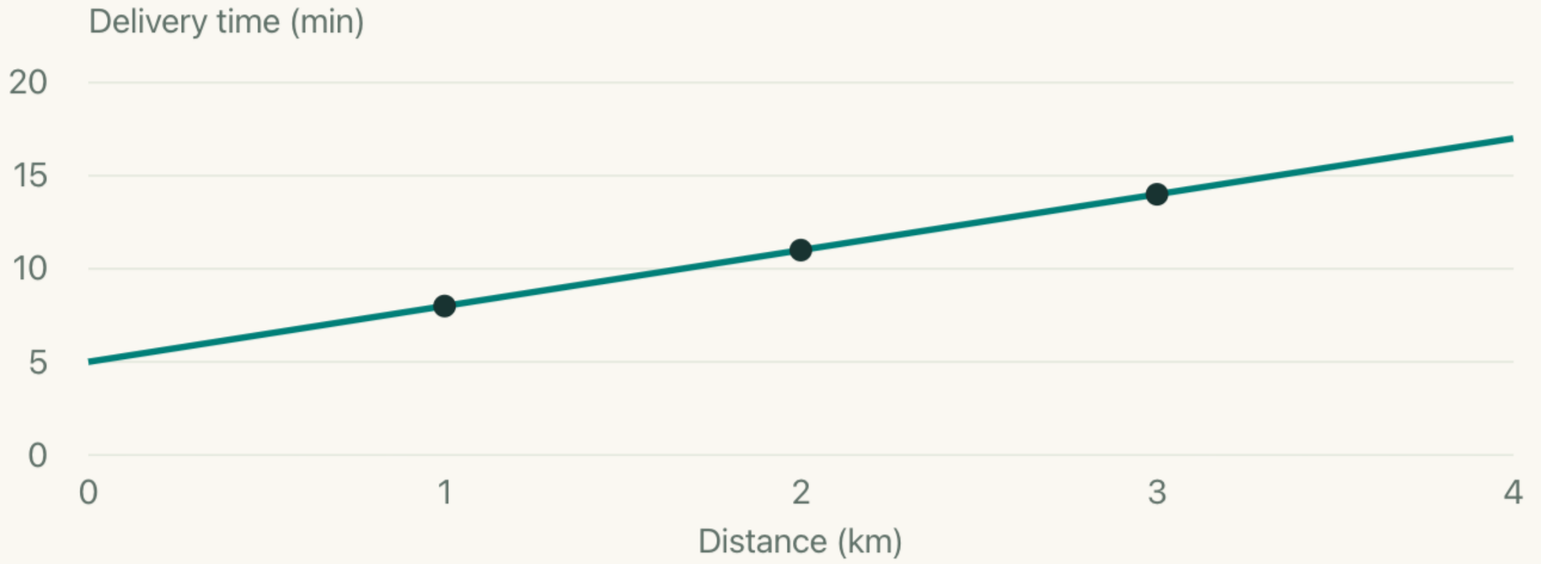
Preparation / bias (minutes) 3

Travel rate / slope (min/km) 2

Loss is calculated across the same three observed deliveries.

Reset parameters

A better fit




```
Estimated minutes = 5 + distance × 3
```

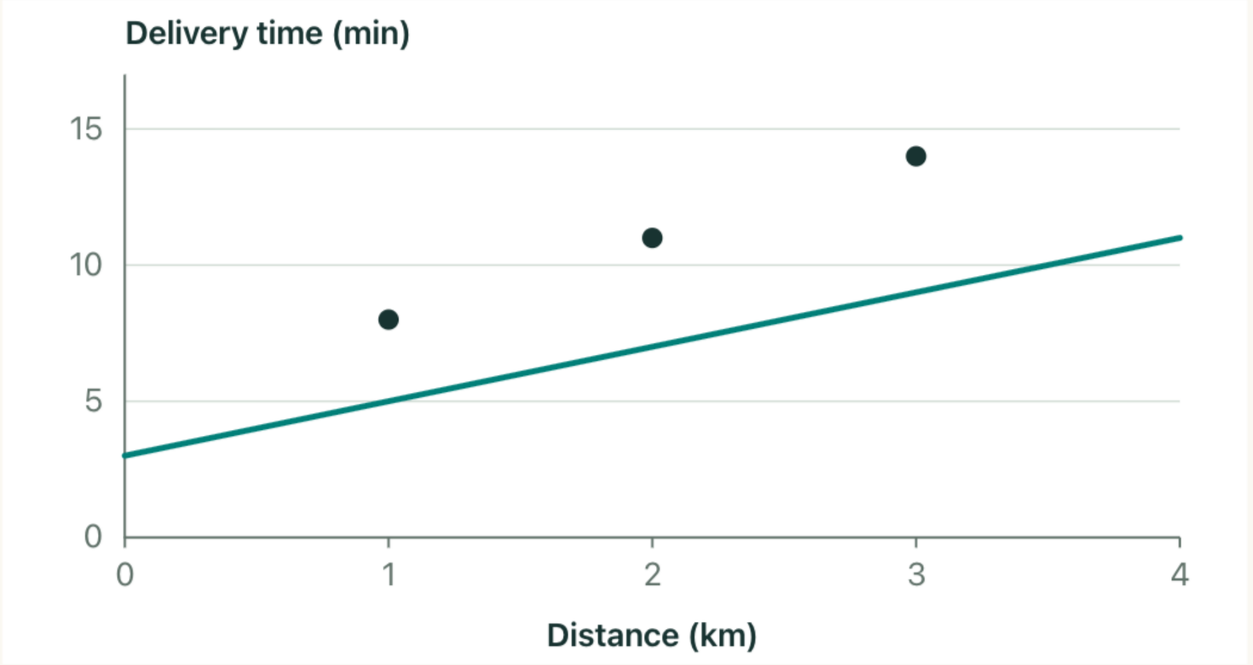
Predictions: 8, 11, 14 minutes

Who chooses the adjustments?

How could the program choose better parameter values?

How can the model choose a better guess?

We found a better fit by moving the sliders.



Predicted minutes = 3 + 2 × distance in km

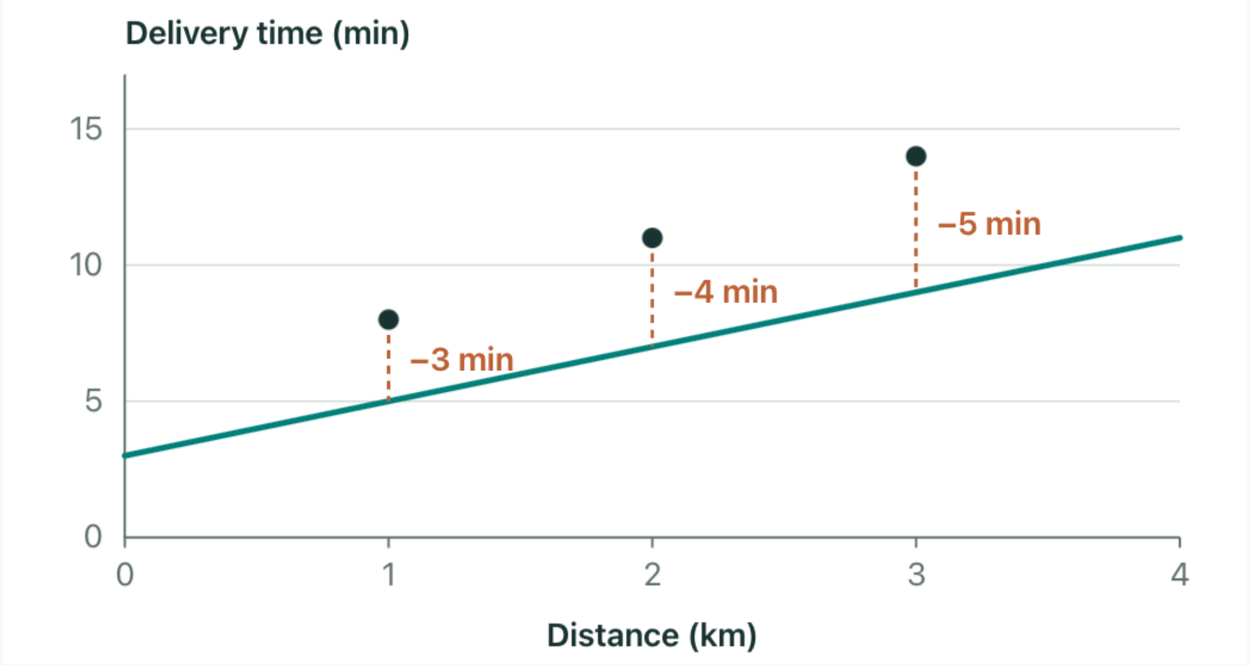
Our first prediction

The dots show the recorded times.

The teal line shows what we predicted.

How can the model choose a better guess?

We found a better fit by moving the sliders.



Predicted minutes = 3 + 2 × distance in km

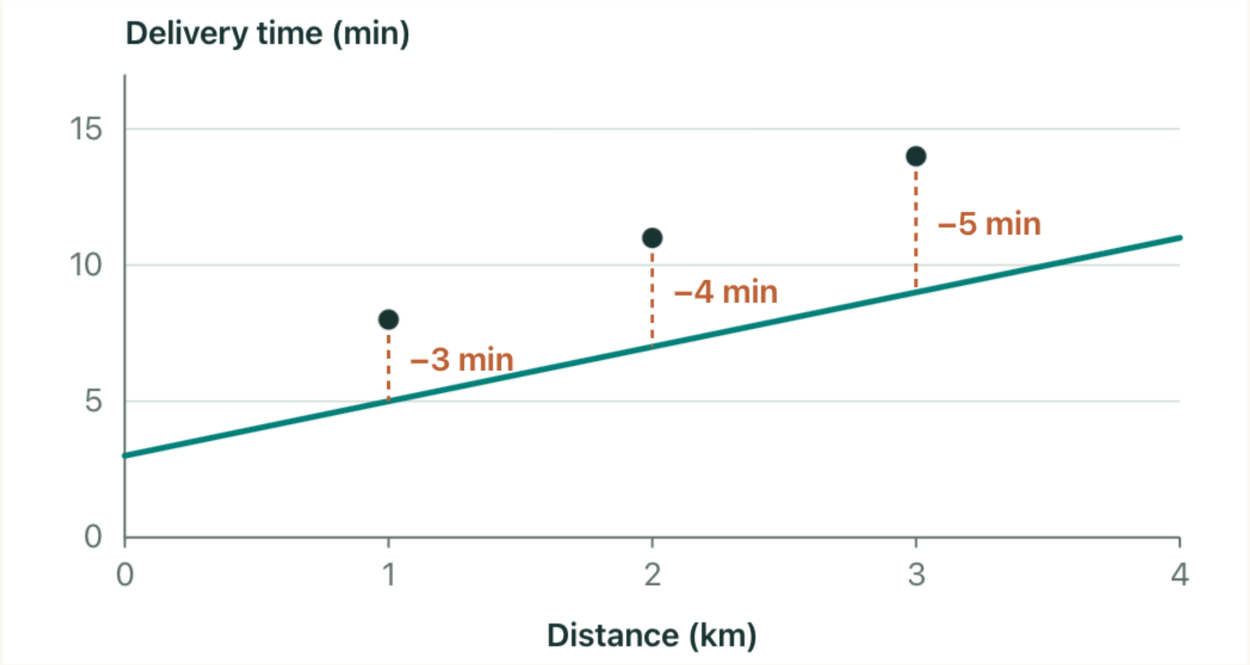
Three prediction errors

We predicted 5, 7, and 9 minutes. The orders took 8, 11, and 14.

Each error is the predicted time minus the recorded time.

How can the model choose a better guess?

We found a better fit by moving the sliders.



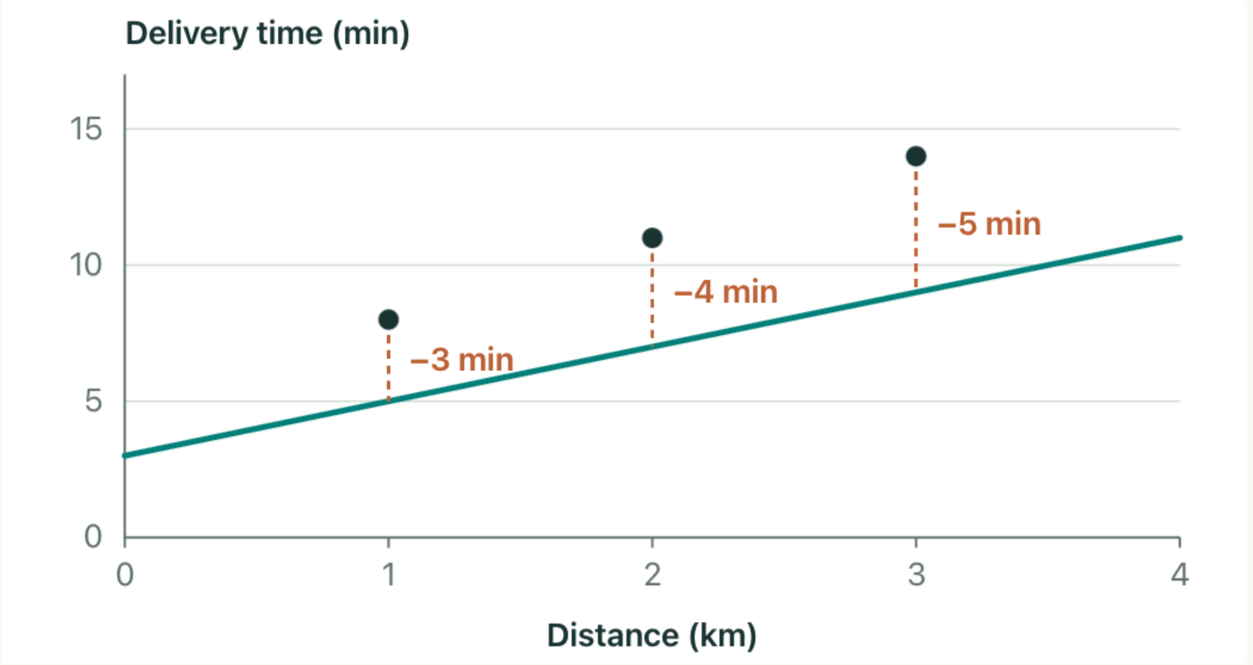
Predicted minutes = 3 + 2 × distance in km

How do we turn three errors into one score?

One score would let us compare different guesses on the same orders.

How can the model choose a better guess?

We found a better fit by moving the sliders.



Predicted minutes = 3 + 2 × distance in km

What if we average the errors?

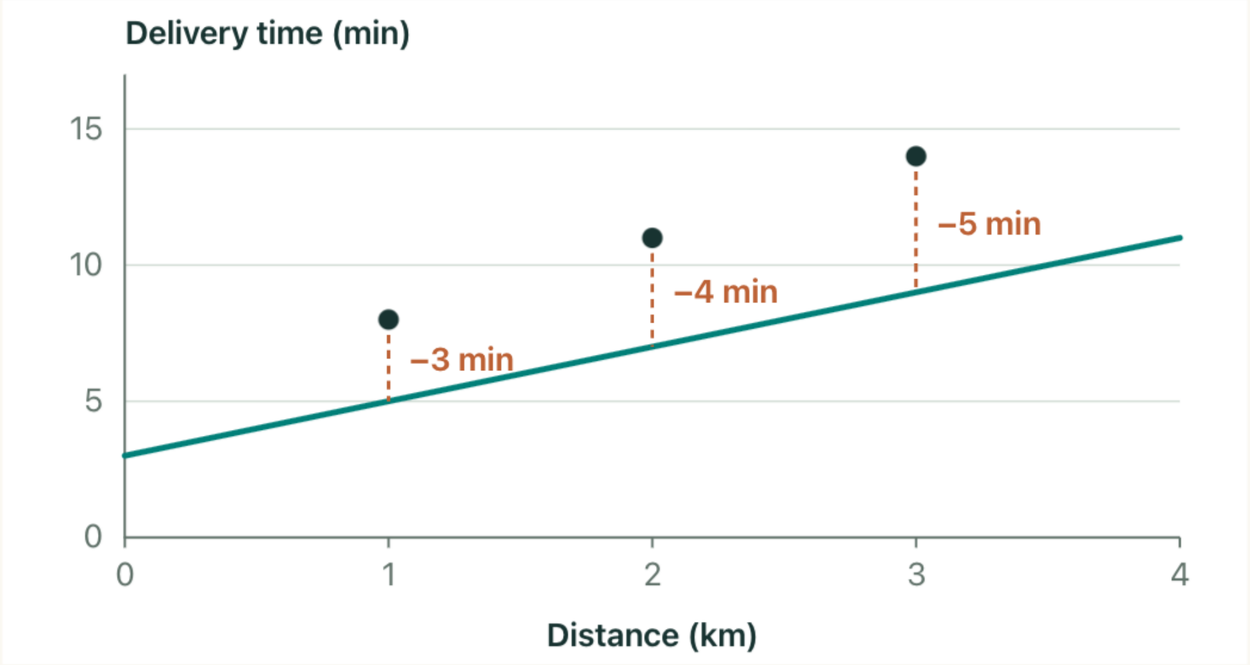
Imagine one prediction is 10 minutes early and another is 10 minutes late.

(−10 + 10) ÷ 2 = **0**

The errors cancel, even though both predictions missed.

How can the model choose a better guess?

We found a better fit by moving the sliders.



Predicted minutes = 3 + 2 × distance in km

Mean squared error

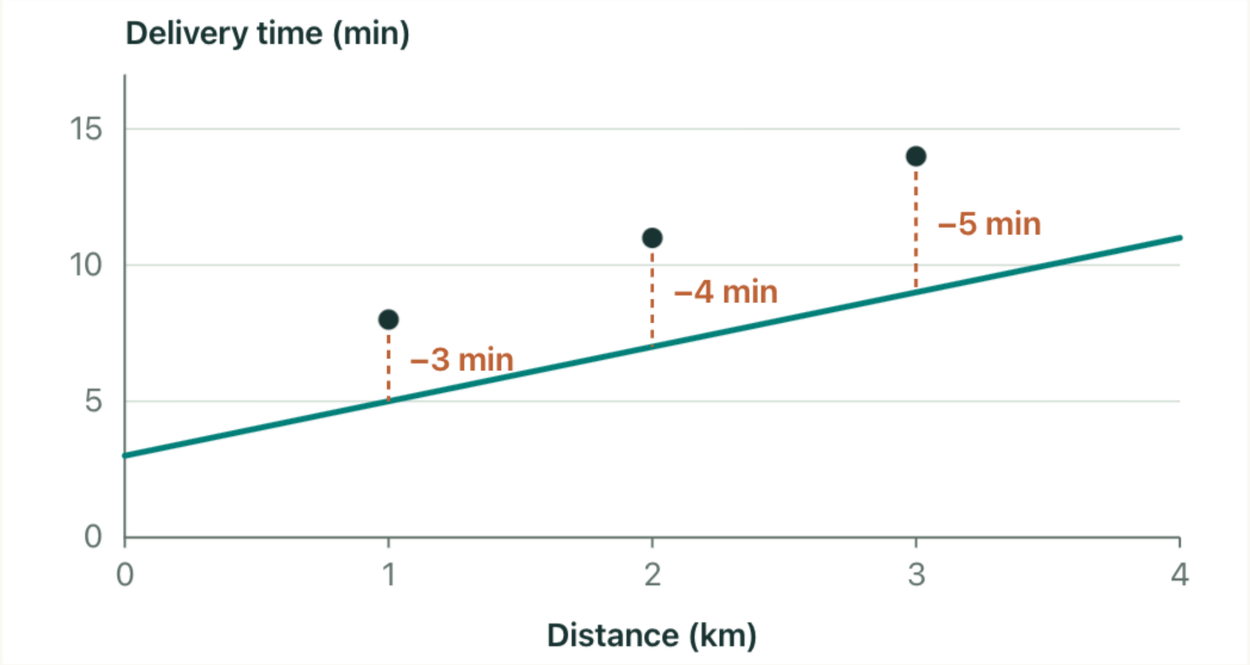
A common way to score numerical predictions.

Square. Every miss adds to the score. A 10-minute miss adds 100, while a 1-minute miss adds 1.

Average. Get one score to compare guesses on the same orders.

How can the model choose a better guess?

We found a better fit by moving the sliders.



Predicted minutes = 3 + 2 × distance in km

Calculate the delivery loss

Square the three errors on the graph and average them.

$[(-3)^2 + (-4)^2 + (-5)^2] \div 3$

$= (9 + 16 + 25) \div 3 \approx$ **16.7 min²**

A loss function for any guess

Predicted delivery time

$$p + rd$$

- ***p*** Preparation time (min)
- ***r*** Travel rate (min/km)
- ***d*** Distance of the order (km)
- ***t*** Recorded delivery time (min)

Three recorded deliveries

Distance (km)	Recorded time (min)
1	8
2	11
3	14

A loss function for any guess

Predicted delivery time

$$p + rd$$

- ***p*** Preparation time (min)
- ***r*** Travel rate (min/km)
- ***d*** Distance of the order (km)
- ***t*** Recorded delivery time (min)

Prediction error

Predicted time minus recorded time

$$p + rd - t$$

Three recorded deliveries

Distance (km)	Recorded time (min)	Prediction error (min)
1	8	
2	11	
3	14	

A loss function for any guess

Predicted delivery time

$$p + rd$$

- p Preparation time (min)
- r Travel rate (min/km)
- d Distance of the order (km)
- t Recorded delivery time (min)

Prediction error

Predicted time minus recorded time

$$p + rd - t$$

Three recorded deliveries

Distance (km)	Recorded time (min)	Prediction error (min)
1	8	$p + r \times 1 - 8$
2	11	$p + r \times 2 - 11$
3	14	$p + r \times 3 - 14$

A loss function for any guess

Predicted delivery time

$$p + rd$$

- ***p*** Preparation time (min)
- ***r*** Travel rate (min/km)
- ***d*** Distance of the order (km)
- ***t*** Recorded delivery time (min)

Prediction error

Predicted time minus recorded time

$$p + rd - t$$

Three recorded deliveries

Start with:

p (preparation time) = **3 minutes**

r (travel rate) = **2 minutes per kilometre**

Distance (km)	Recorded time (min)	Prediction error (min)
1	8	3 + 2 × 1 – 8 = **–3**
2	11	3 + 2 × 2 – 11 = **–4**
3	14	3 + 2 × 3 – 14 = **–5**

A loss function for any guess

Predicted delivery time

$$p + rd$$

- ***p*** Preparation time (min)
- ***r*** Travel rate (min/km)
- ***d*** Distance of the order (km)
- ***t*** Recorded delivery time (min)

Prediction error

Predicted time minus recorded time

$$p + rd - t$$

One loss for three orders

Use each row's expression with p and r. Square the three errors and average them.

$$L\left(p, r\right) = \frac{\left(p + r - 8\right)^2 + \left(p + 2r - 11\right)^2 + \left(p + 3r - 14\right)^2}{3}$$

Each pair of guesses gives one score.

A loss function for any guess

Predicted delivery time

$$p + rd$$

- ***p*** Preparation time (min)
- ***r*** Travel rate (min/km)
- ***d*** Distance of the order (km)
- ***t*** Recorded delivery time (min)

Prediction error

Predicted time minus recorded time

$$p + rd - t$$

Preparation stays at 3 minutes

Substitute $p = 3$ in each error. Now the score depends on the travel rate alone.

$$L\left(3, r\right) = \frac{\left(r - 5\right)^2 + \left(2r - 8\right)^2 + \left(3r - 11\right)^2}{3}$$

A loss function for any guess

Predicted delivery time

$$p + rd$$

- ***p*** Preparation time (min)
- ***r*** Travel rate (min/km)
- ***d*** Distance of the order (km)
- ***t*** Recorded delivery time (min)

Prediction error

Predicted time minus recorded time

$$p + rd - t$$

Simplify to get the curve's equation

Expand the squares and collect like terms.

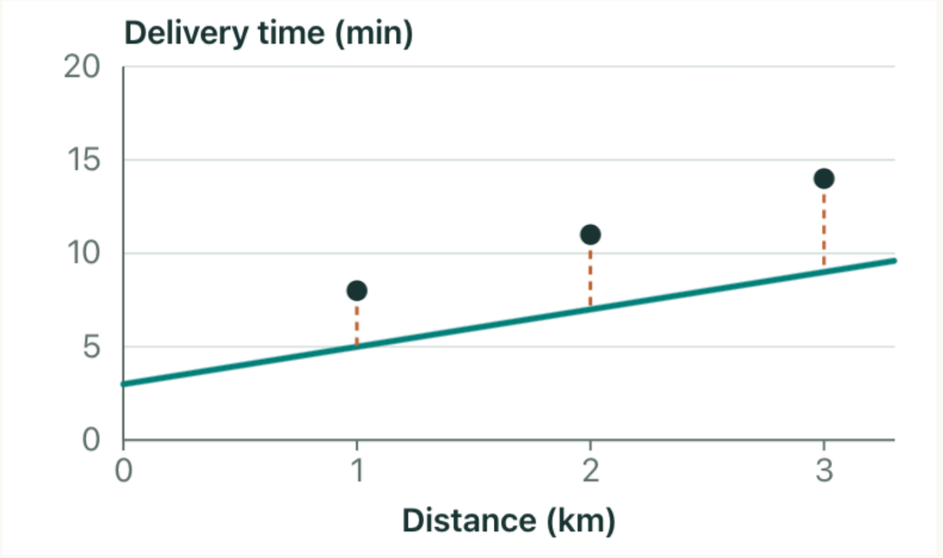
$$L\left(3, r\right) = \frac{14}{3}r^2 - 36r + 70$$

At $r = 2$, the loss is **16.7 min²**. The same score we calculated from the three errors.

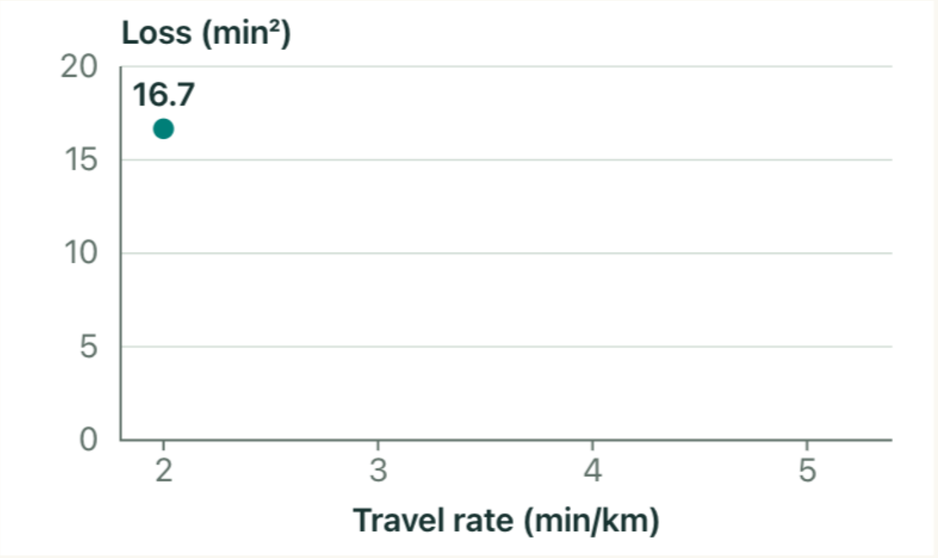
The r^2 term gives the graph its quadratic shape.

How does loss change with travel rate?

Preparation stays fixed at **3 minutes**. Each travel rate gives one loss score.



● Recorded deliveries — Prediction line

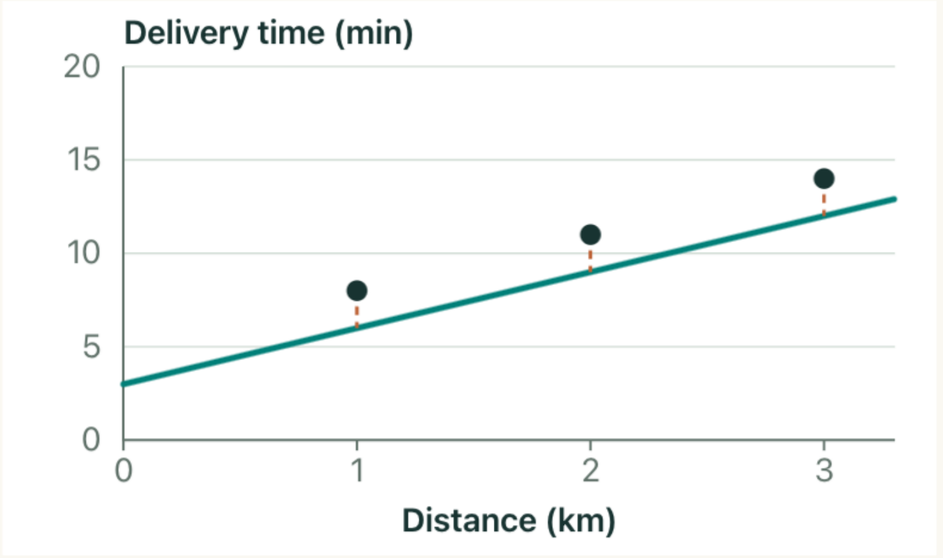


Each dot scores one travel-rate guess.

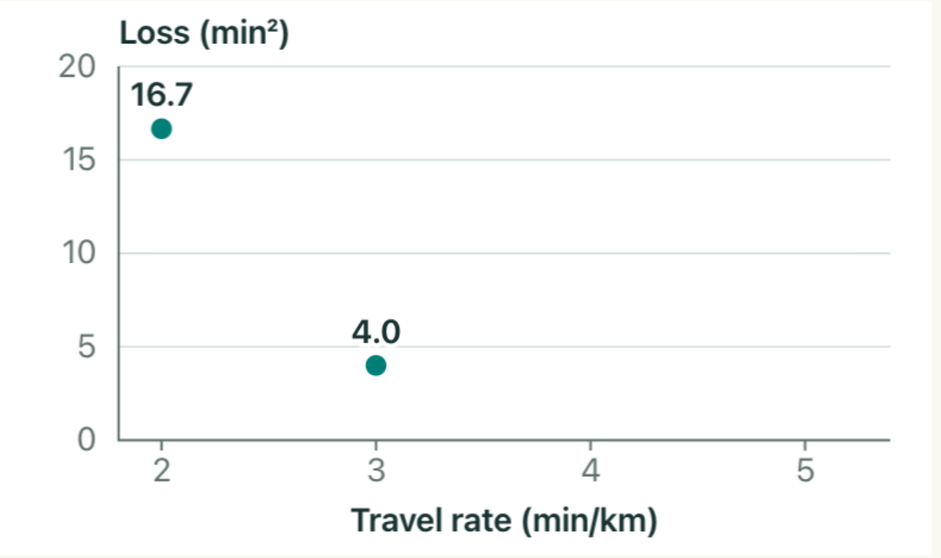
2 min/km. Our first guess predicts times that are too short.

How does loss change with travel rate?

Preparation stays fixed at **3 minutes**. Each travel rate gives one loss score.



● Recorded deliveries — Prediction line

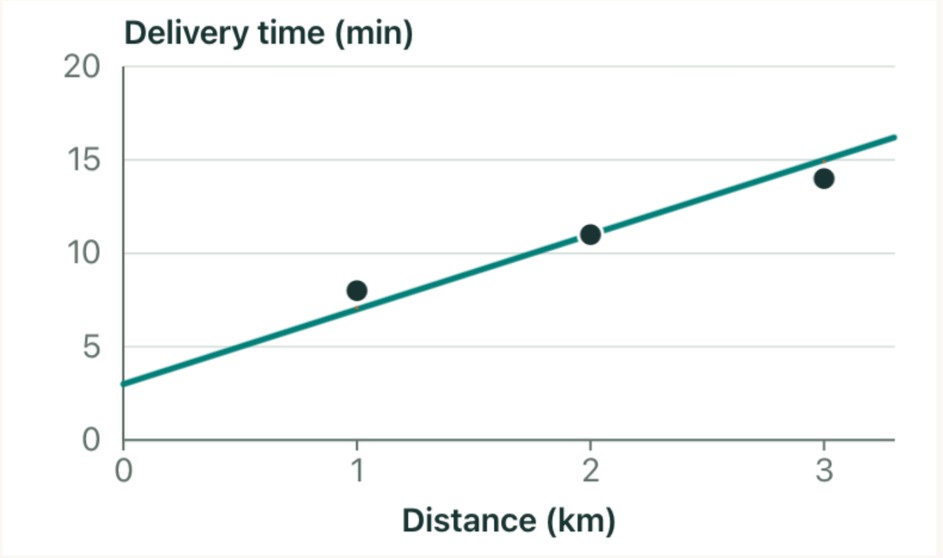


Each dot scores one travel-rate guess.

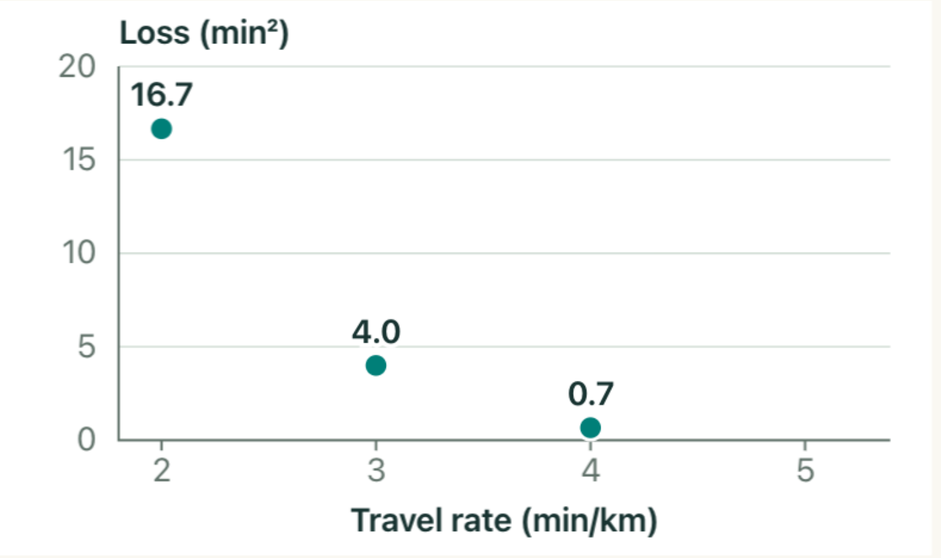
3 min/km. The steeper line brings the predictions closer to the recorded times.

How does loss change with travel rate?

Preparation stays fixed at **3 minutes**. Each travel rate gives one loss score.



● Recorded deliveries — Prediction line

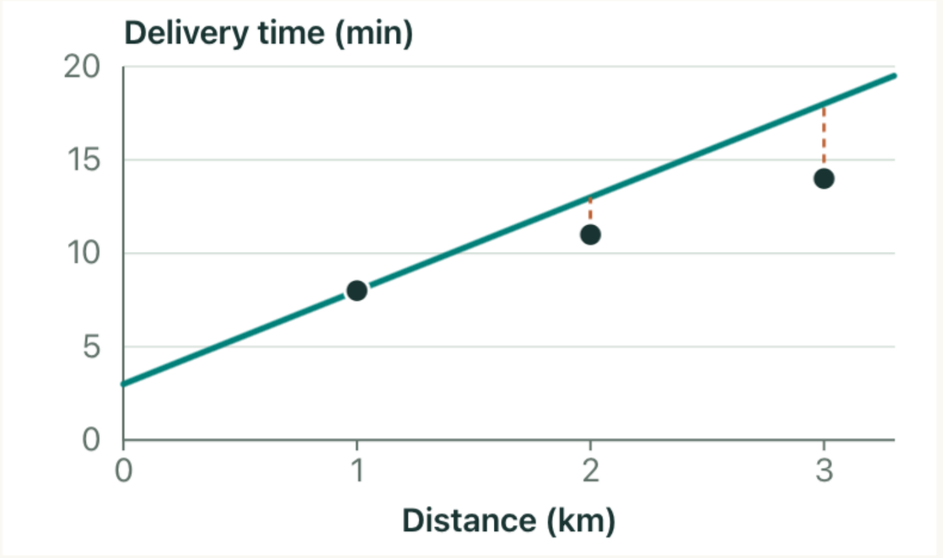


Each dot scores one travel-rate guess.

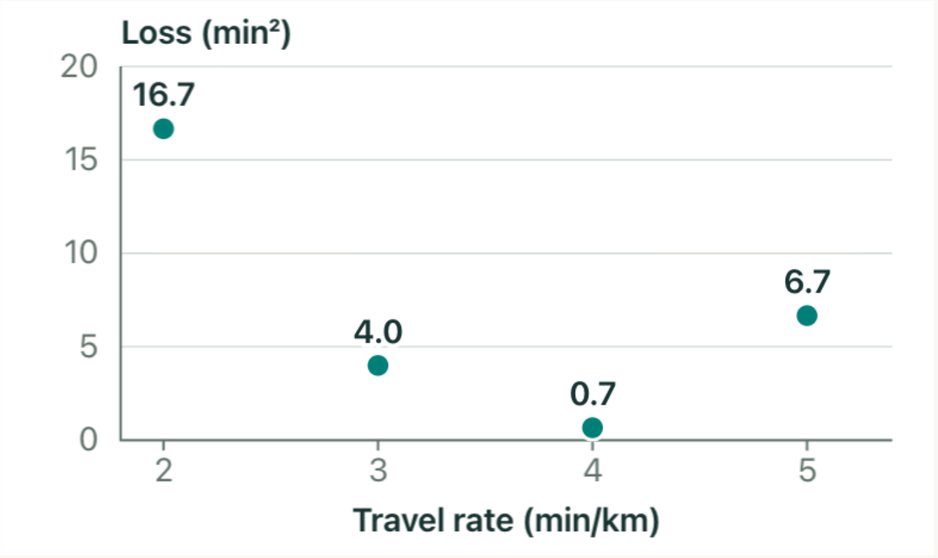
4 min/km. The line fits the three orders better overall.

How does loss change with travel rate?

Preparation stays fixed at **3 minutes**. Each travel rate gives one loss score.



● Recorded deliveries — Prediction line

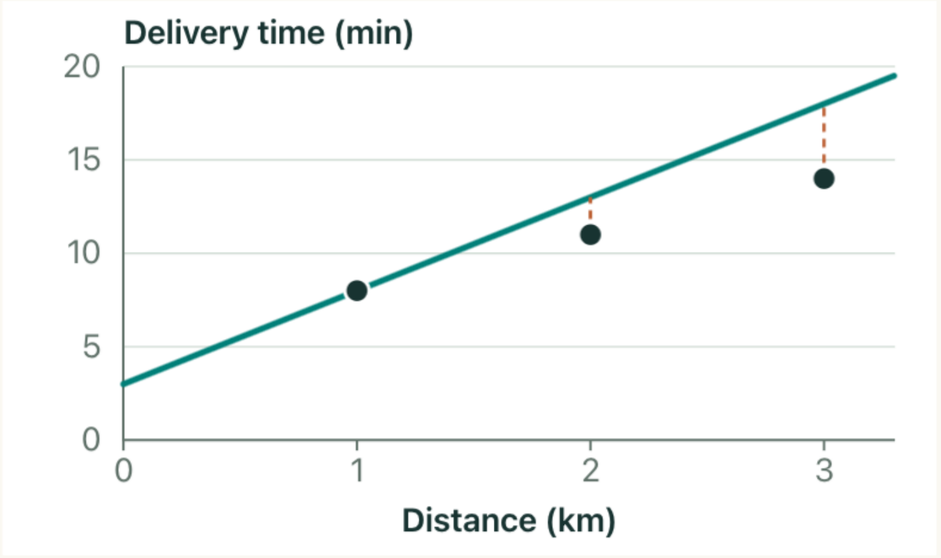


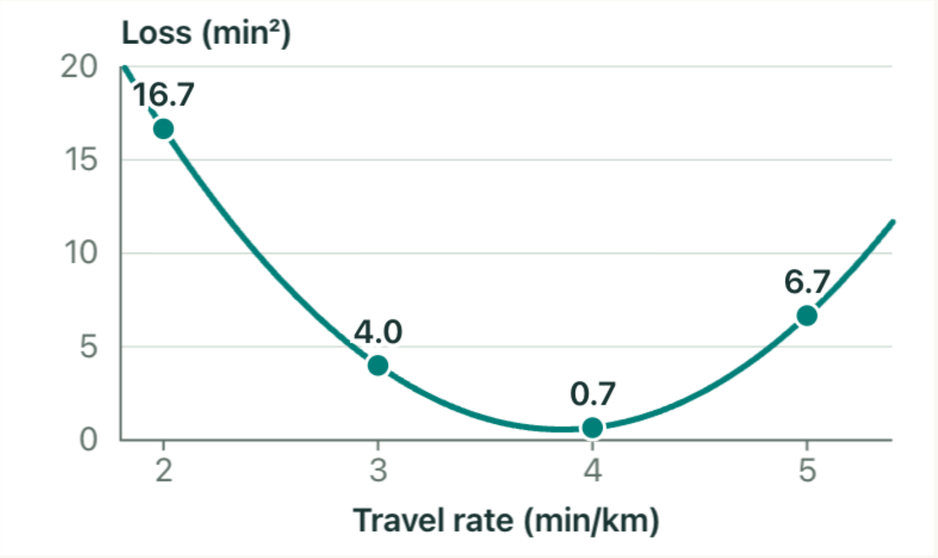
Each dot scores one travel-rate guess.

5 min/km. The longer-distance predictions overshoot, and loss rises.

How does loss change with travel rate?

Preparation stays fixed at **3 minutes**. Each travel rate gives one loss score.





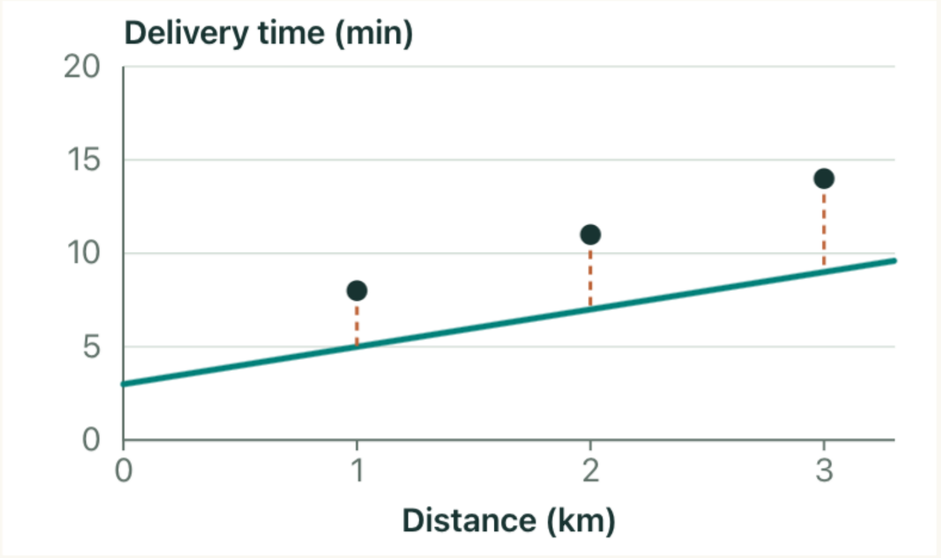
● Recorded deliveries — Prediction line

Each dot scores one travel-rate guess.

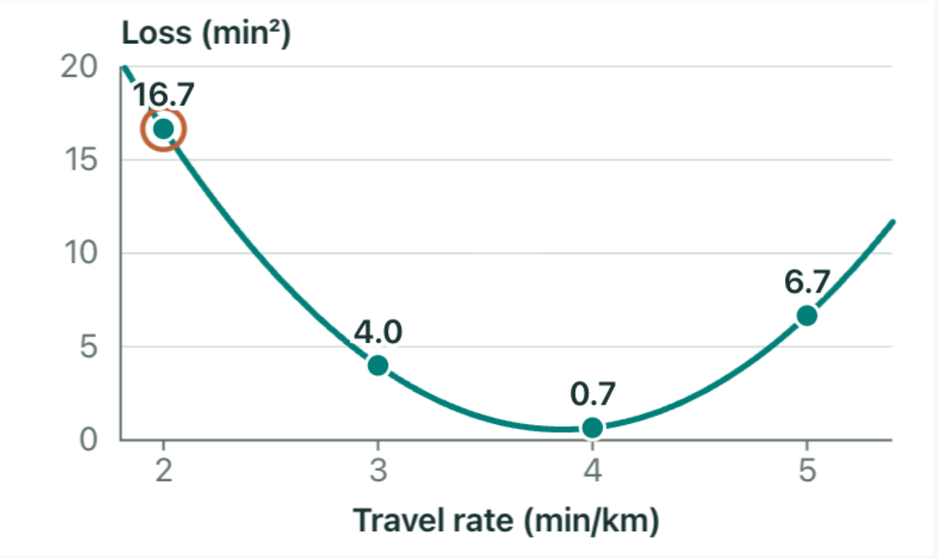
Every point on the curve scores a whole prediction line against the same three orders.

How does loss change with travel rate?

Preparation stays fixed at **3 minutes**. Each travel rate gives one loss score.



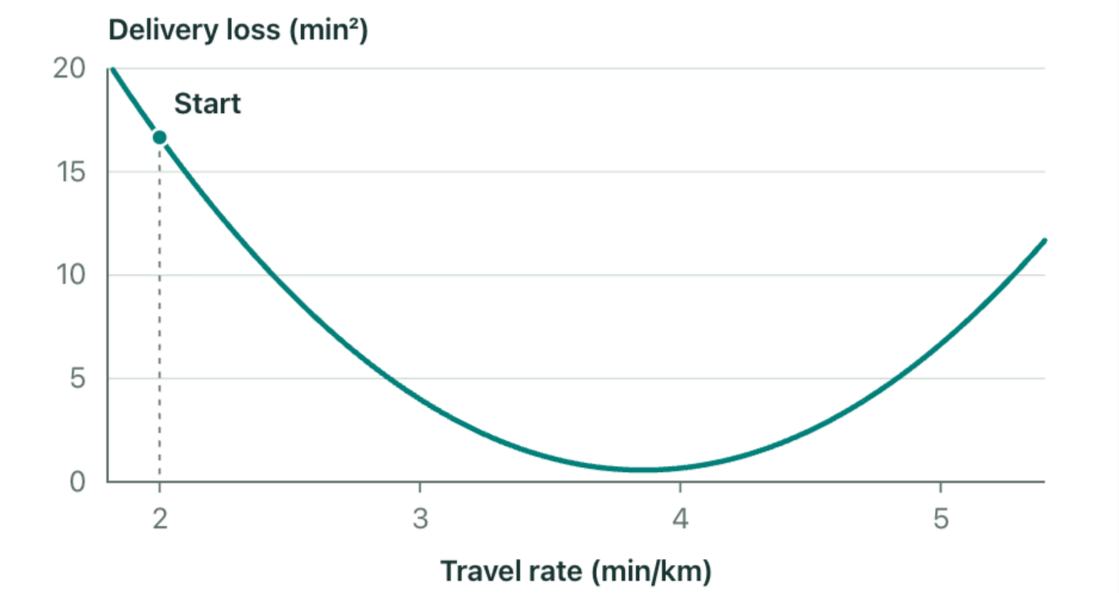
● Recorded deliveries — Prediction line



Each dot scores one travel-rate guess.

Back at 2 min/km. Which way should the model move to lower the loss?

What does a derivative measure?

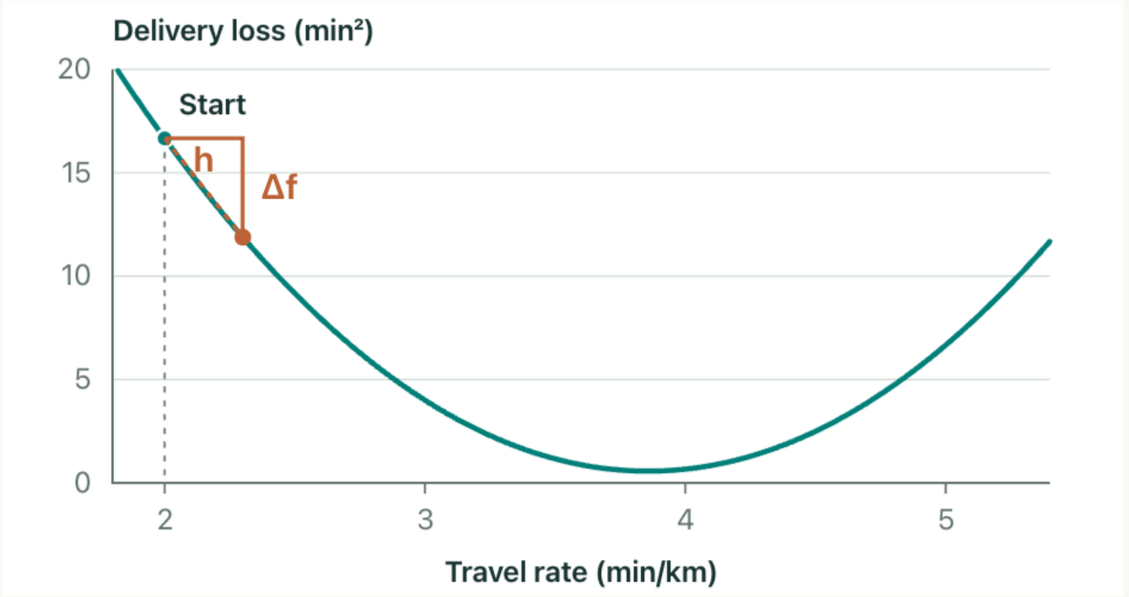


Preparation stays fixed at 3 minutes.

Start at 2 min/km

The loss is **16.7 min²**. If we increase the rate a little, how fast does the loss change?

What does a derivative measure?



Preparation stays fixed at 3 minutes.

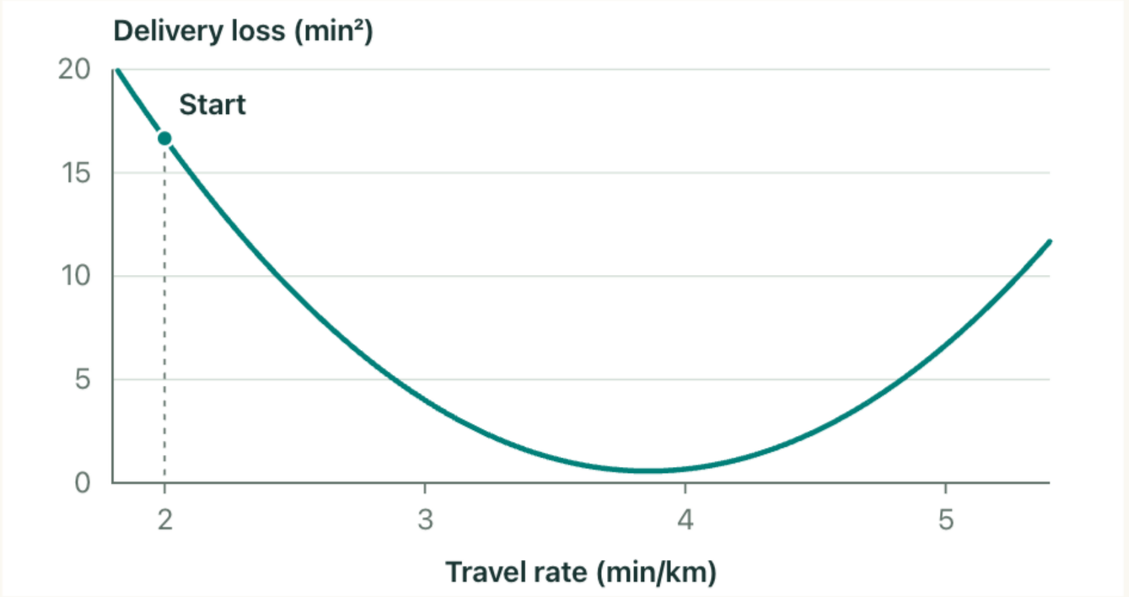
The definition of a derivative

Compare the function at x and at $x + h$.

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Divide the vertical change Δf by the horizontal change h. Let h approach zero to get the slope at x.

What does a derivative measure?



Preparation stays fixed at 3 minutes.

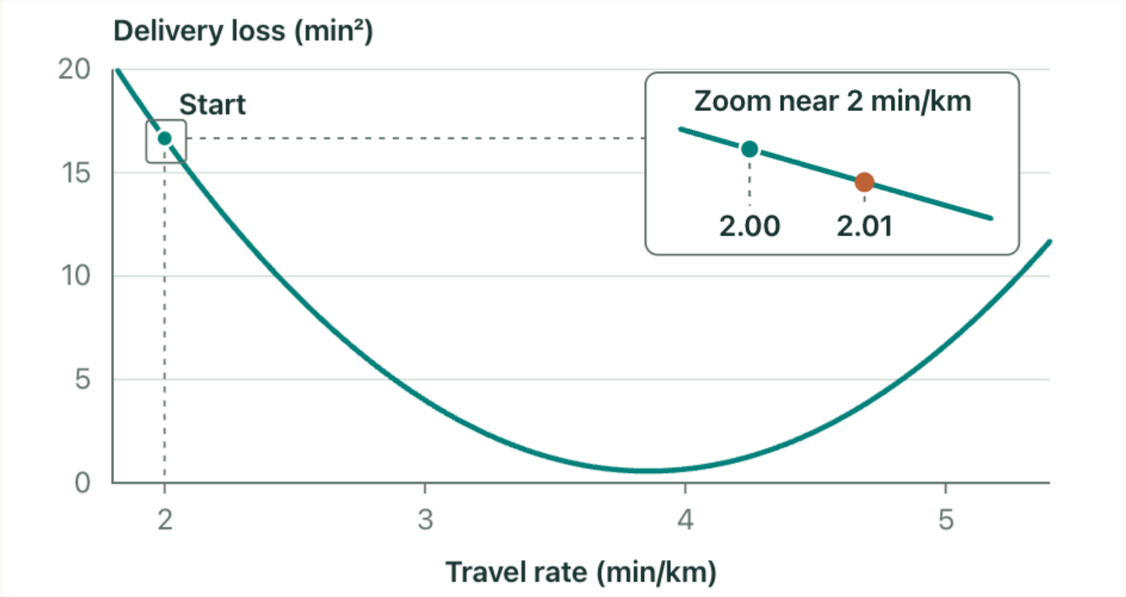
The derivative at 2 min/km

Use $L(3, r)$ as the function and set the travel rate r to 2.

$$\lim_{h \to 0} \frac{L(3, 2+h) - L(3, 2)}{h}$$

As h approaches zero, this gives the exact rate of change of loss with respect to travel rate.

What does a derivative measure?



Preparation stays fixed at 3 minutes.

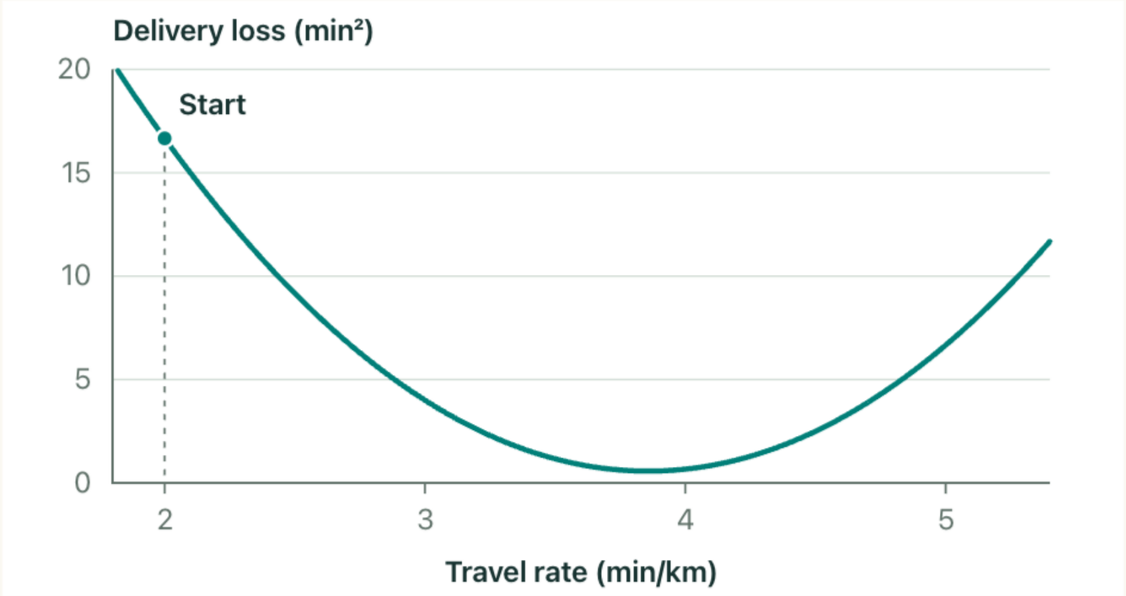
Try $h = 0.01$

The second point is at **2.01 min/km**.

$$\frac{L(3, 2.01) - L(3, 2)}{0.01}$$

≈ −0.17287 ÷ 0.01 ≈ **−17.29**

What does a derivative measure?



Preparation stays fixed at 3 minutes.

Differentiate the loss equation

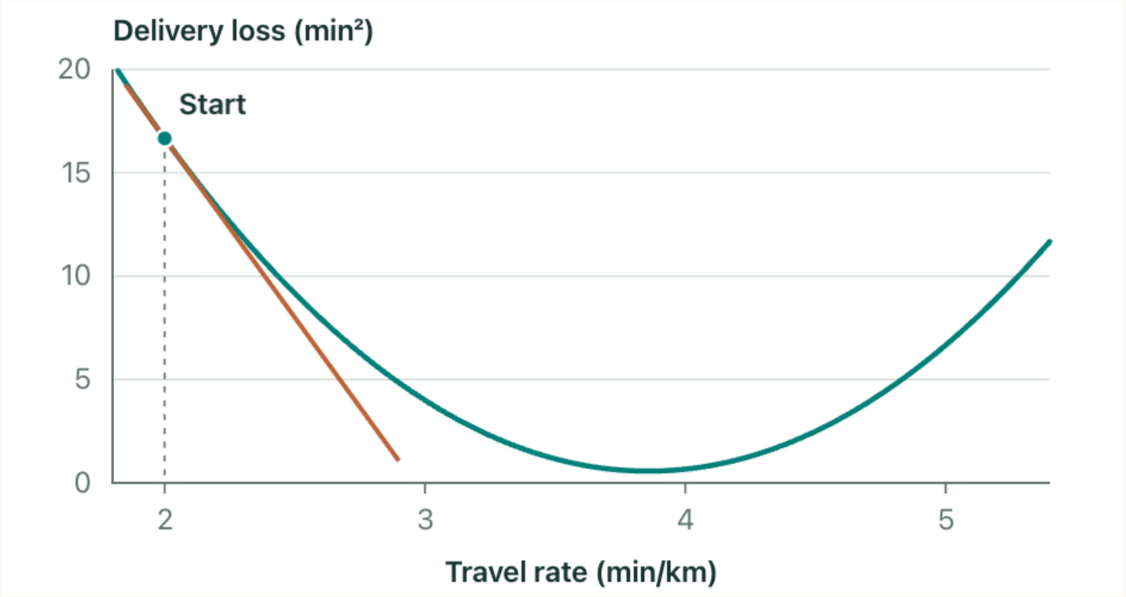
$$L\left(3, r\right) = \frac{14}{3}r^2 - 36r + 70$$

Differentiate each term with respect to r. The derivatives of r^2, r, and 70 are $2r$, 1, and 0.

$$\frac{\mathrm{d}}{\mathrm{d}r}L = \frac{14}{3} \cdot 2r - 36 \cdot 1 + 0$$

$$= \frac{28}{3}r - 36$$

What does a derivative measure?



Preparation stays fixed at 3 minutes.

Compare with the exact slope

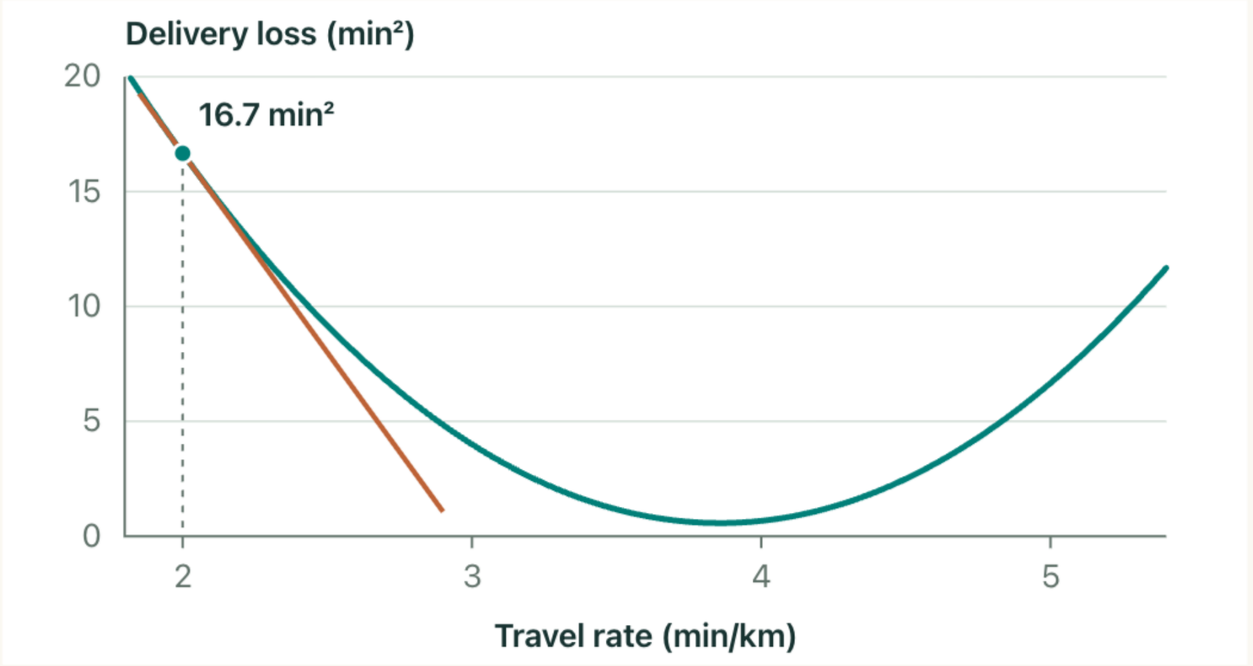
Plug $r = 2$ into the derivative to get the slope at the starting point.

$$\begin{aligned} &\frac{28}{3} \cdot 2 - 36 \\ &= \frac{56}{3} - 36 \\ &= \frac{-52}{3} \end{aligned}$$

Estimate	Exact (rounded)
−17.29	**−17.33**

The negative slope points downhill to the right. A small increase in travel rate lowers the loss here.

One step toward lower loss

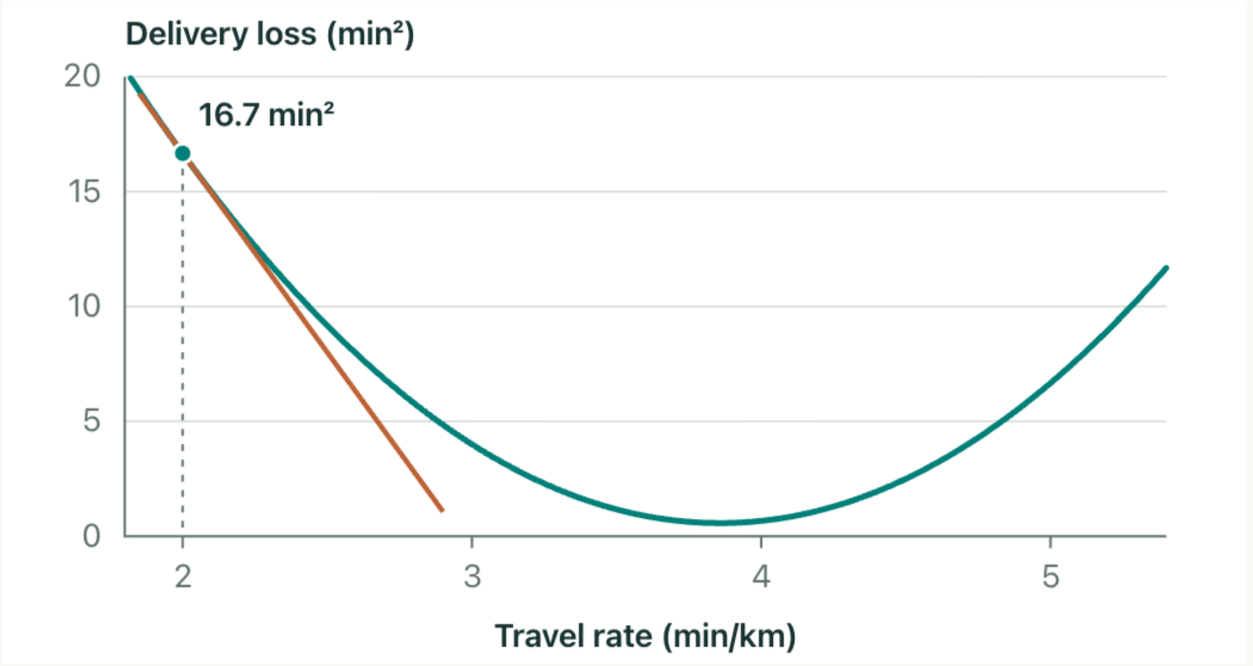


Preparation stays fixed at 3 minutes.

How far should we move?

At **2 min/km**, the slope is **−52/3**. It points downhill to the right.

One step toward lower loss



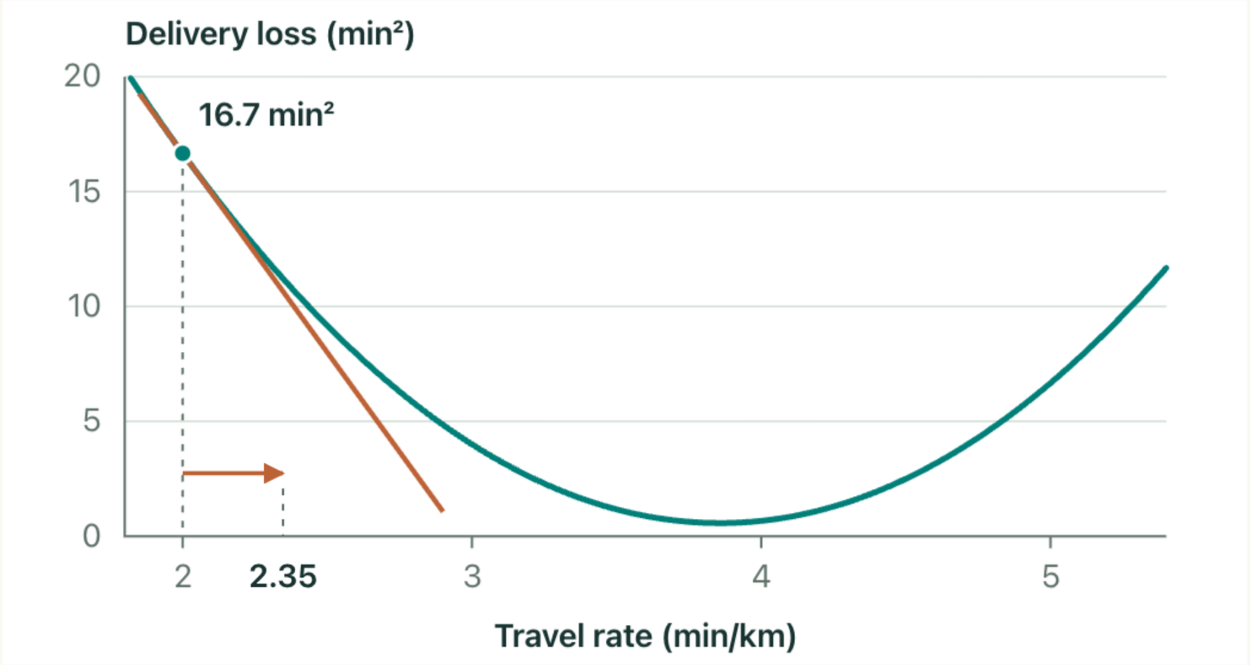
Preparation stays fixed at 3 minutes.

The update rule

New rate = current rate
– learning rate × slope

The learning rate scales the step. Subtracting a negative slope increases the travel rate.

One step toward lower loss



Preparation stays fixed at 3 minutes.

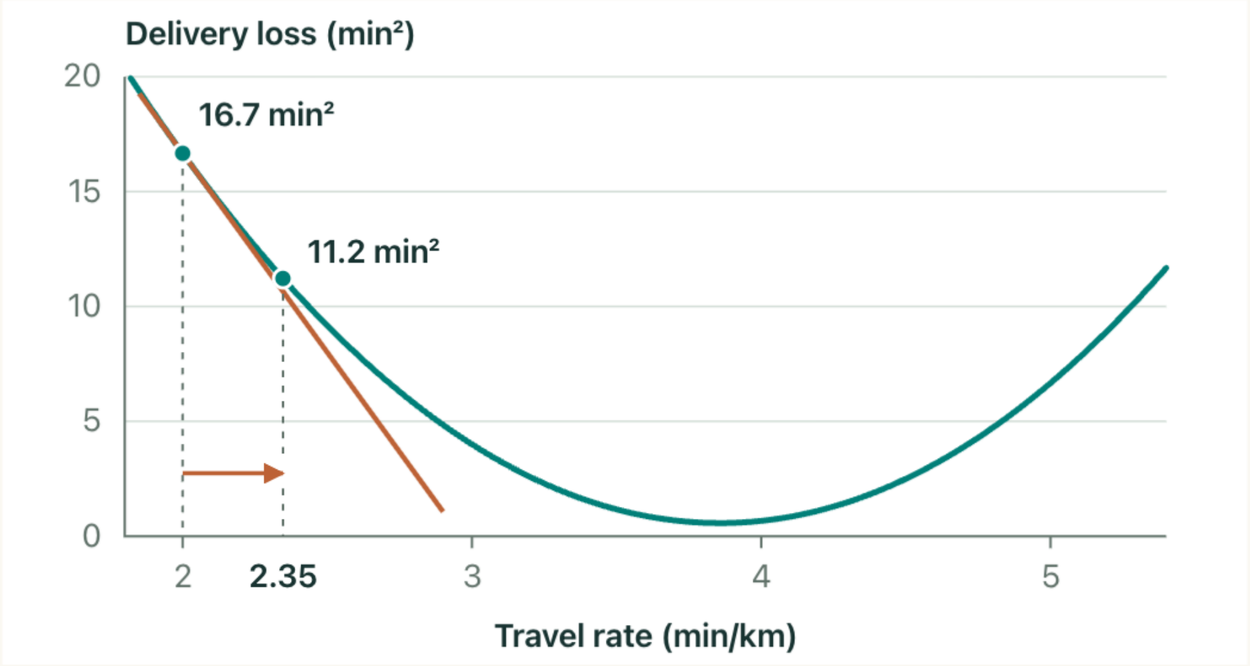
Take one step

Use a learning rate of **0.02**.

$2 - 0.02 \times (-52/3)$

$\approx$ **2.35 min/km**

One step toward lower loss



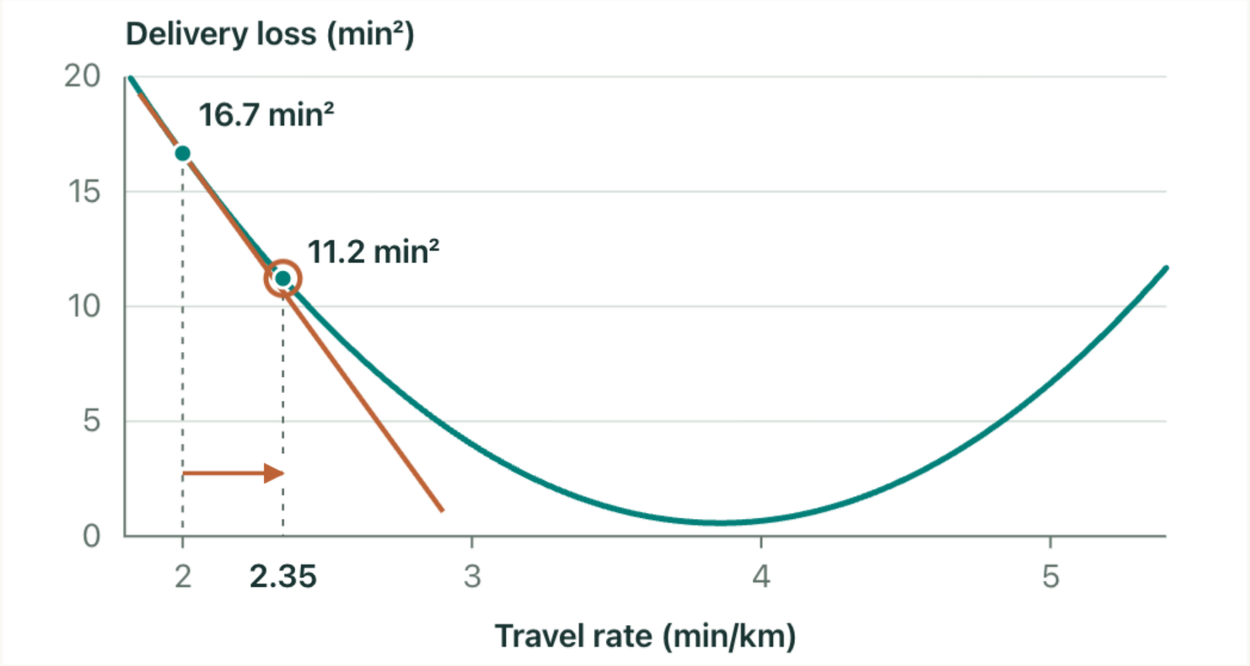
Preparation stays fixed at 3 minutes.

Check the new loss

Use the updated rate to score the same three deliveries.

16.7 → **11.2 min²**

One step toward lower loss

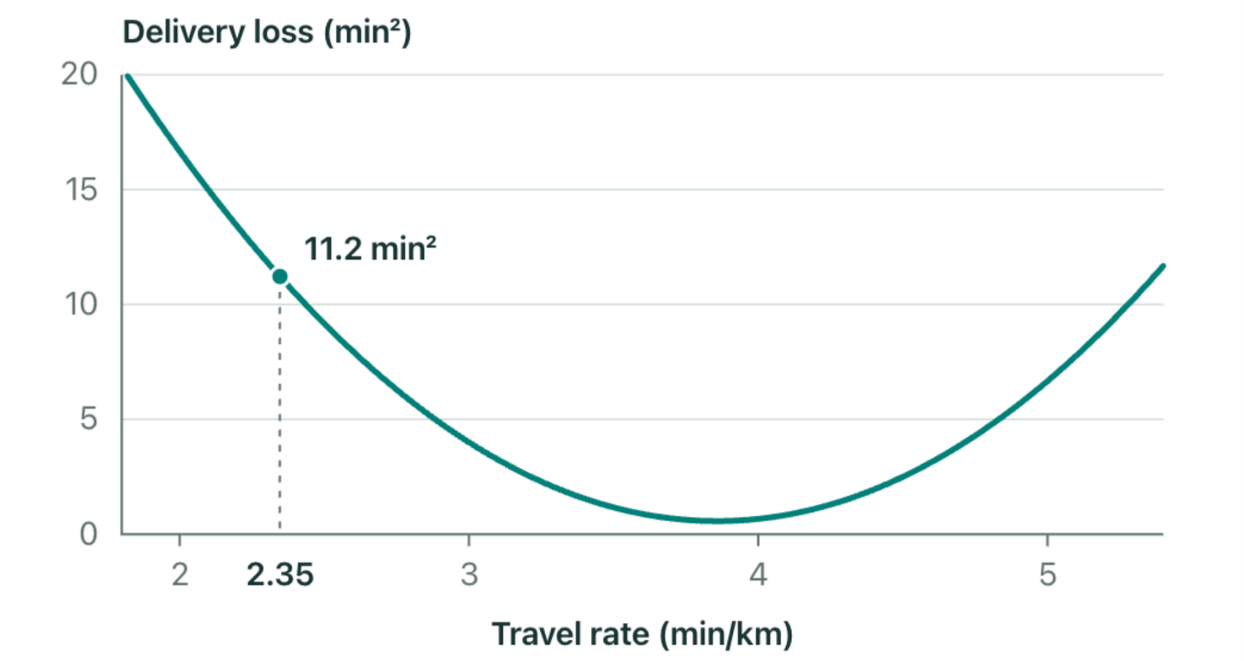


Preparation stays fixed at 3 minutes.

Gradient descent

Find the slope at the current guess, take a small downhill step, and repeat from the new guess.

What happens if we keep going?



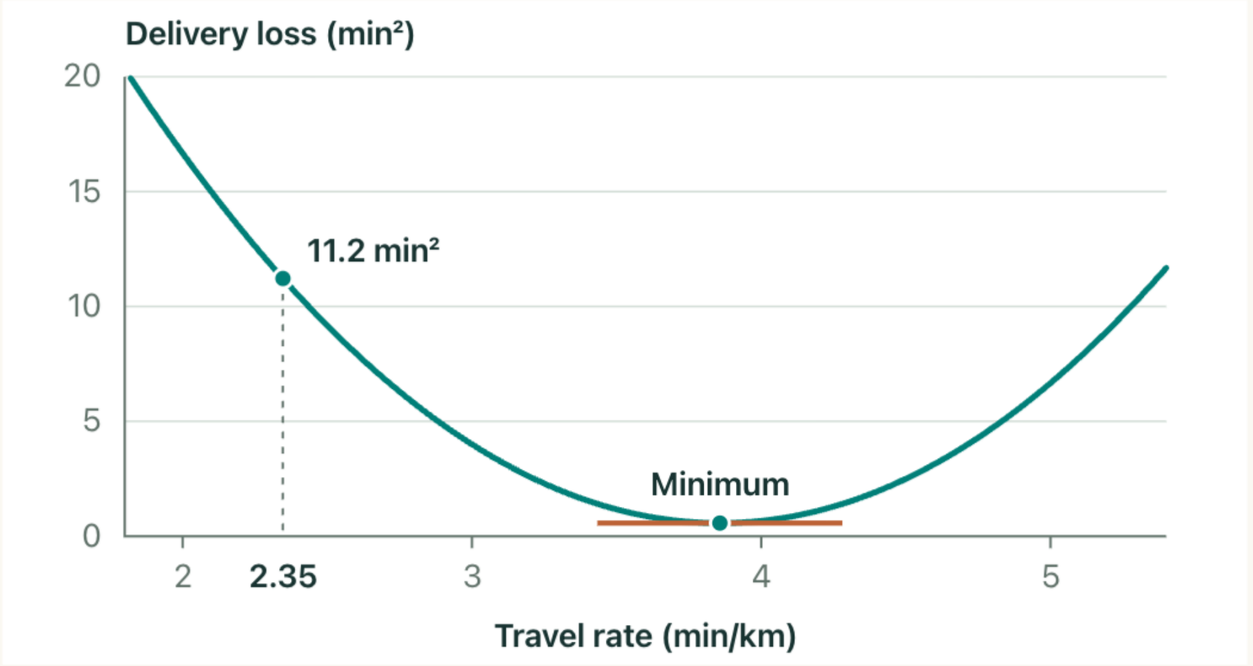
Preparation stays fixed at 3 minutes.

After one step

We reached **2.35 min/km** and brought loss down to **11.2 min²**.

What happens if we keep repeating the same update?

What happens if we keep going?



Preparation stays fixed at 3 minutes.

The slope flattens

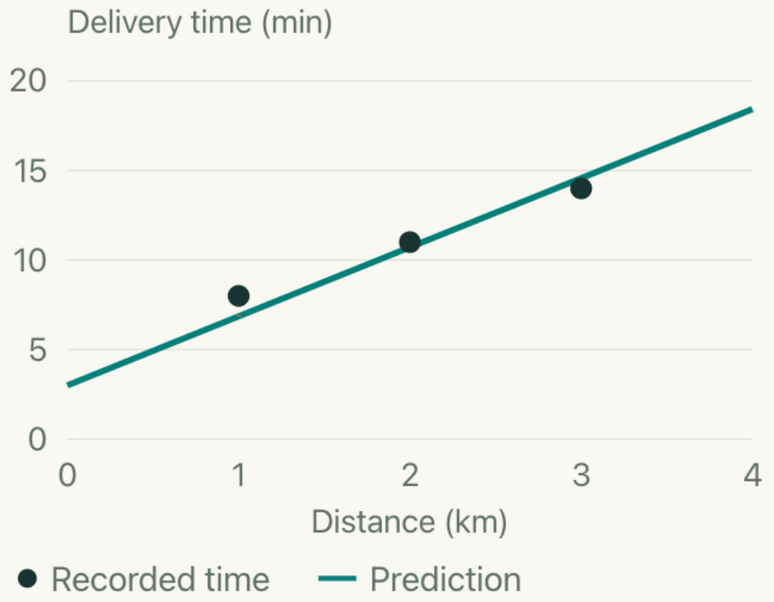
Repeating the update takes us closer to the bottom. A flatter slope gives a smaller step at the same learning rate.

At **3.86 min/km**, loss is **0.57 min²**. The tangent is flat, so the slope and the update are both zero.

Could changing preparation time lower the loss further?

What if we change preparation time?

Keep the travel rate at **27/7 ≈ 3.86 min/km**, the rate that minimized loss with 3 minutes of preparation.



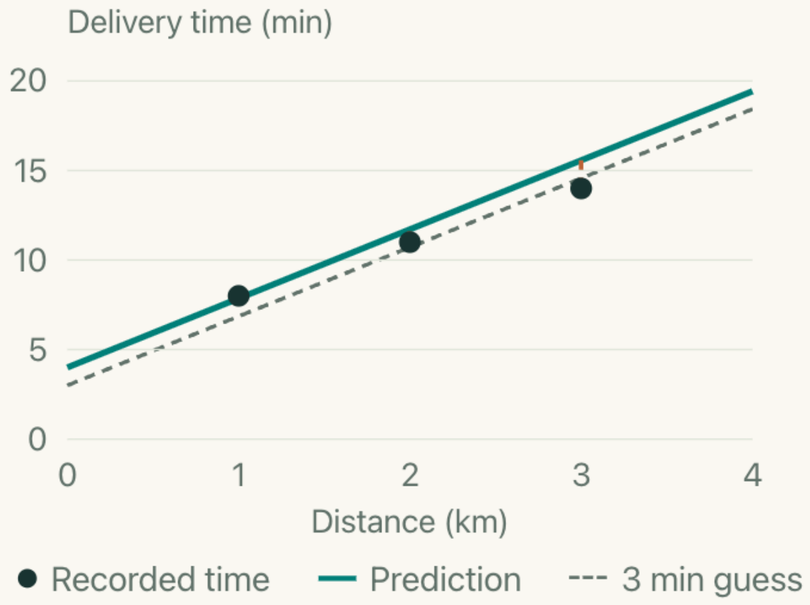
Preparation time = 3 min

Distance (km)	Recorded time (min)	Prediction error (min)
1	8	$3 + (27/7) \times 1 - 8 \approx \mathbf{-1.14}$
2	11	$3 + (27/7) \times 2 - 11 \approx \mathbf{-0.29}$
3	14	$3 + (27/7) \times 3 - 14 \approx \mathbf{+0.57}$

Mean squared error ≈ **0.57 min²**

What if we change preparation time?

Keep the travel rate at **27/7 ≈ 3.86 min/km**, the rate that minimized loss with 3 minutes of preparation.



Preparation time = 4 min

Distance (km)	Recorded time (min)	Prediction error (min)
1	8	4 + (27/7) × 1 – 8 ≈ **–0.14**
2	11	4 + (27/7) × 2 – 11 ≈ **+0.71**
3	14	4 + (27/7) × 3 – 14 ≈ **+1.57**

Each error rises by 1 min. Mean squared error = **1.00 min²**.

What if we change preparation time?

Build the loss from the three deliveries

Use p in place of the preparation-time guess. For the 1 km delivery, the error becomes $p + 27/7 - 8 = p - 29/7$.

Square and average the three errors to get the loss for any p.

$$L\left(p, \frac{27}{7}\right) = \frac{\left(p - \frac{29}{7}\right)^2 + \left(p - \frac{23}{7}\right)^2 + \left(p - \frac{17}{7}\right)^2}{3}$$

What if we change preparation time?

Simplify the quadratic

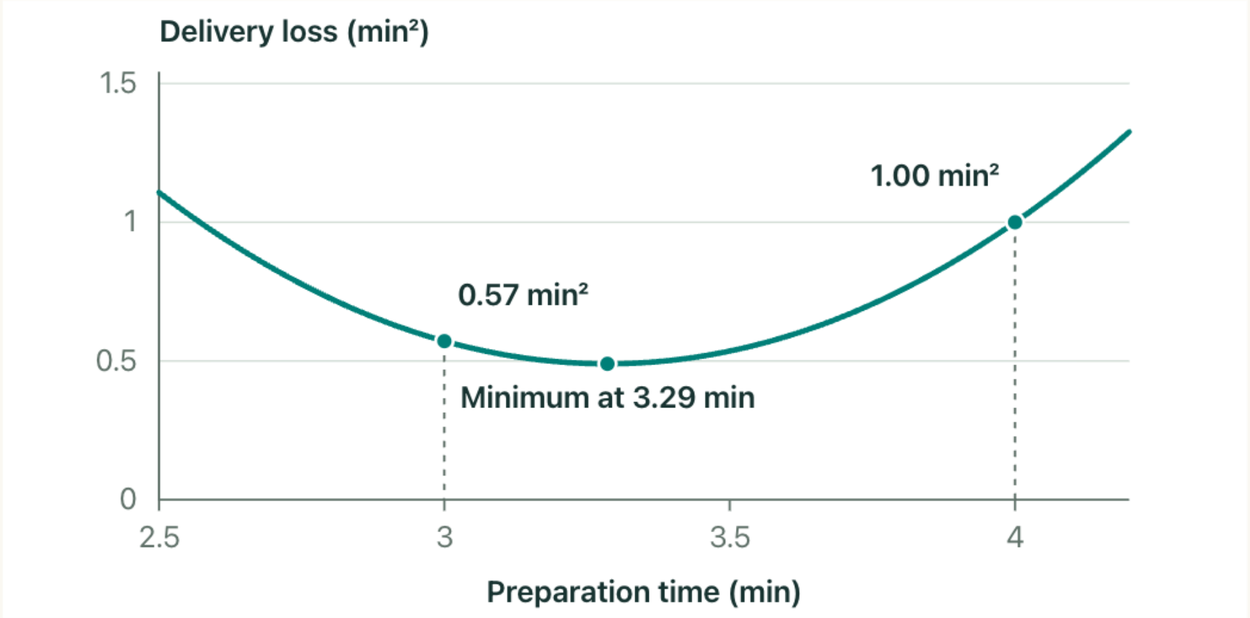
Expand the three squares, combine like terms, and group them into one square.

$$L\left(p, \frac{27}{7}\right) = \left(p - \frac{23}{7}\right)^2 + \frac{24}{49}$$

The squared term reaches zero at $p = 23/7$, so the lowest point is at about **3.29 minutes**.

What if we change preparation time?

$$L\left(p, \frac{27}{7}\right) = \left(p - \frac{23}{7}\right)^2 + \frac{24}{49}$$



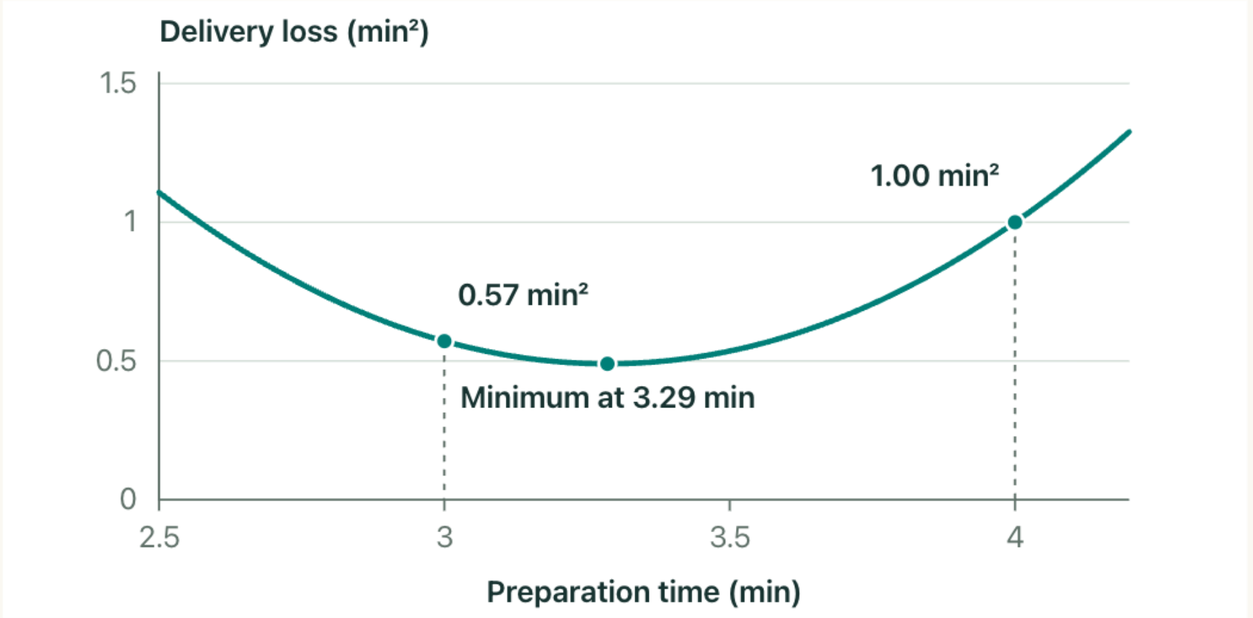
Travel rate stays fixed at approximately 3.86 min/km.

Plot loss against preparation time

The two guesses give **0.57 min²** at 3 minutes and **1.00 min²** at 4 minutes. The minimum lies between them at **3.29 minutes**.

What if we change preparation time?

$$L\left(p, \frac{27}{7}\right) = \left(p - \frac{23}{7}\right)^2 + \frac{24}{49}$$



Travel rate stays fixed at approximately 3.86 min/km.

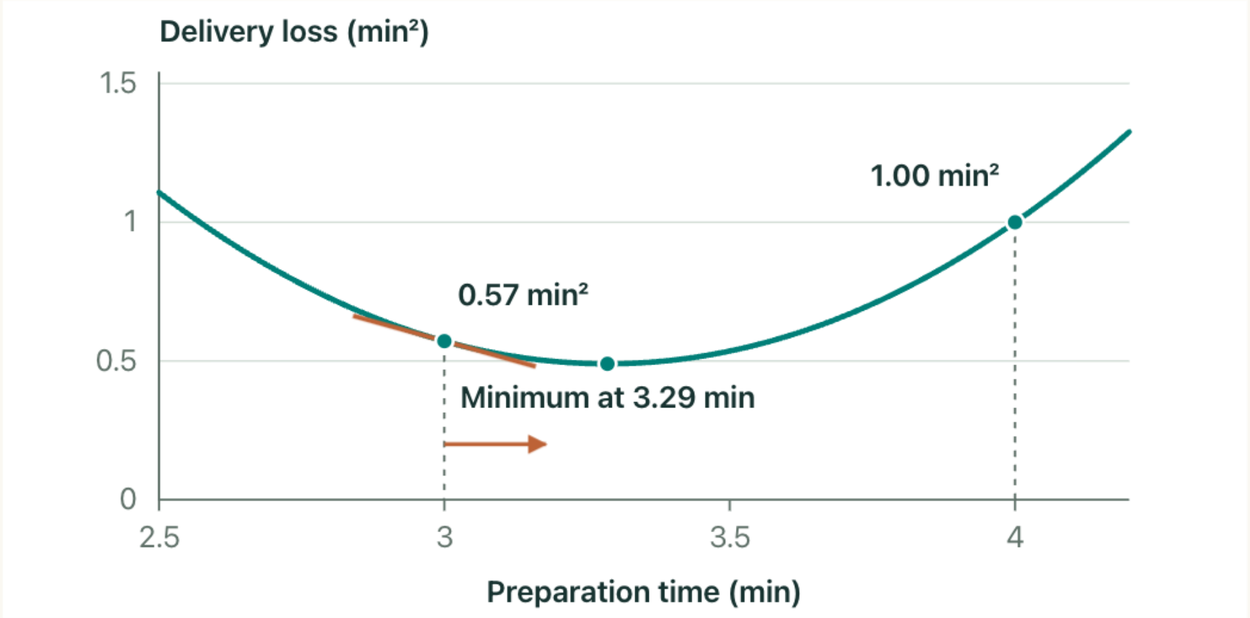
Differentiate with respect to prep time

A **partial derivative** measures how loss changes as we vary preparation time and hold the travel rate fixed.

$$\frac{\partial L}{\partial p} = 2\left(p - \frac{23}{7}\right)$$

What if we change preparation time?

$$L\left(p, \frac{27}{7}\right) = \left(p - \frac{23}{7}\right)^2 + \frac{24}{49}$$



Travel rate stays fixed at approximately 3.86 min/km.

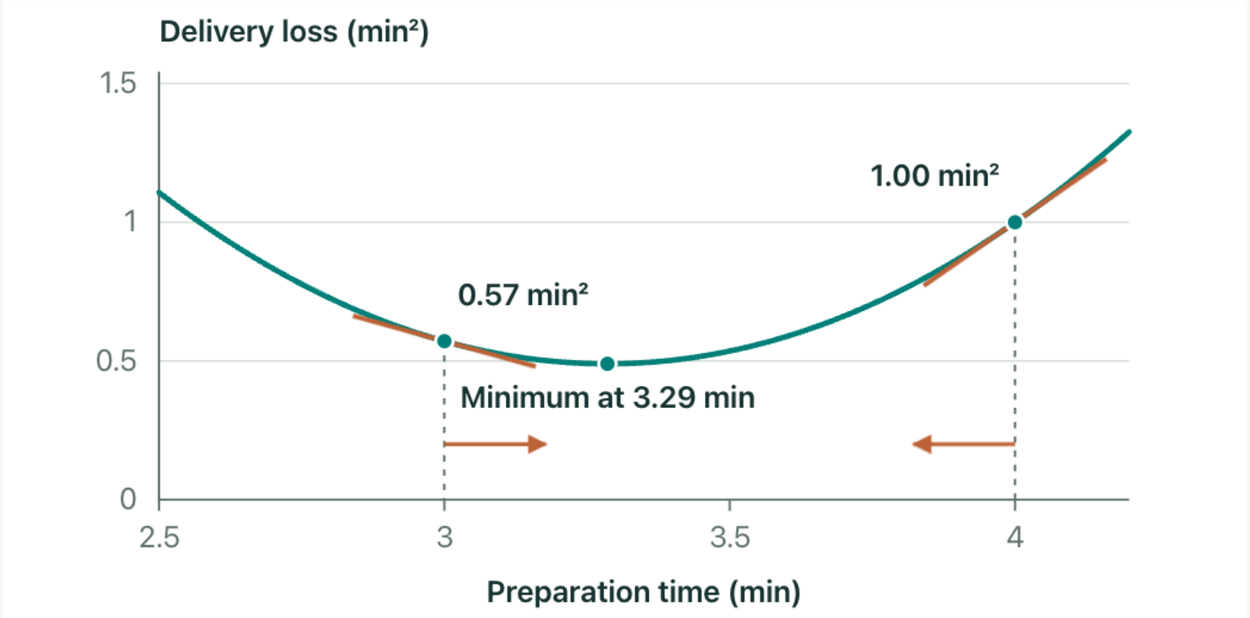
At 3 minutes, move right

2(3 – 23/7) = **–4/7 ≈ –0.57**

The negative slope means a small increase in preparation time lowers loss.

What if we change preparation time?

$$L\left(p, \frac{27}{7}\right) = \left(p - \frac{23}{7}\right)^2 + \frac{24}{49}$$



Travel rate stays fixed at approximately 3.86 min/km.

At 4 minutes, move left

2(4 – 23/7) = **+10/7 ≈ +1.43**

The slope is positive here and zero at the minimum, around **3.29 minutes**.

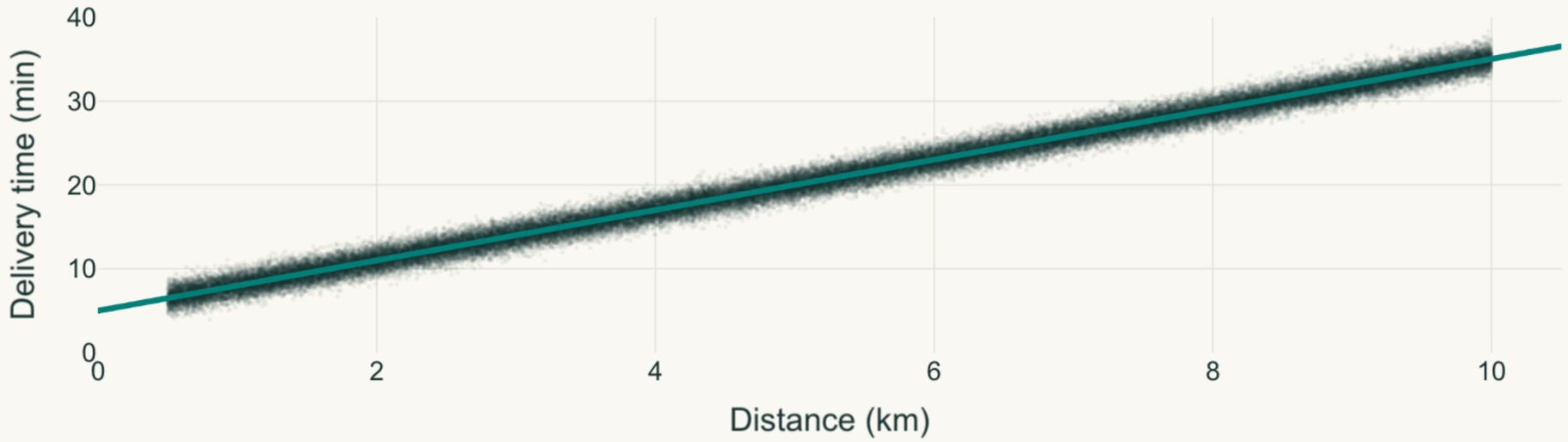
Back at 3 minutes, the travel-rate slope was zero and the prep-time slope was negative. How could we use both slopes in a training update?

Learning from a stream of deliveries

Model: Linear regression · Optimizer: Stochastic gradient descent (SGD) · Learning rate: 0.008

100,000 deliveries

Loss on fixed check sample: **0.822**



Each incoming batch gives the model another gradient update.

Complete

Run / Pause | Reset simulation

Starts again from our original guesses: 3 min and 2 min/km

Predicting numbers or categories

Regression

Predict a number

Example: delivery time

11 minutes

Predicting numbers or categories

Regression

Predict a number

Example: delivery time

11 minutes

Classification

Predict a category

Example: delivery status

On time or late